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NiceJob Website

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ABSTRACT (ARABIC)

في سوق العمل الذي يتطور بسرعة اليوم، أصبح البحث عن عمل مهمة شاقة بشكل متزايد للأفراد من مختلف الخلفيات والمؤهلات. يمكن أن تكون عملية العثور على وظيفة مناسبة مرهقة وتستغرق وقتاً طويلاً، وتتطلب فهماً شاملاً لديناميكيات سوق العمل واستراتيجيات فعالة للبحث عن وظيفة. يهدف هذا المشروع إلى تطوير منصة للبحث عن عمل تسمى "NiceJop" لتسهيل عملية البحث عن عمل لكل من الباحثين عن عمل وأصحاب العمل. ستوفر المنصة واجهة سهلة الاستخدام مع إمكانيات بحث متقدمة، مما يسمح للمستخدمين بالعثور بسهولة على فرص العمل ذات الصلة. سيتمكن الباحثون عن عمل من إنشاء ملفات تعريف شخصية، وتحميل سيرتهم الذاتية، والتقدم للوظائف من خلال المنصة. ومن ناحية أخرى، سيتمكن أصحاب العمل من الوصول إلى لوحة تحكم شاملة حيث يمكنهم نشر قوائم الوظائف ومراجعة الطلبات وإدارة عملية التوظيف بكفاءة. سيستخدم المشروع منهجيات تطوير سريعة لضمان المرونة والقدرة على التكيف مع المتطلبات المتغيرة. سيتم إنشاء المنصة باستخدام تقنيات الويب الحديثة، بما في ذلك أطر الواجهة الأمامية لواجهة مستخدم سريعة الاستجابة وبديهية، وبنية خلفية قابلة للتطوير للتعامل مع قاعدة المستخدمين المتنامية. بشكل عام، تهدف "NiceJop" إلى إحداث ثورة في تجربة البحث عن عمل من خلال ربط الباحثين عن عمل بفرص العمل المناسبة وتبسيط عملية التوظيف لأصحاب العمل.



ABSTRACT (ENGLISH)

In today's rapidly evolving job market, the employment search has become an increasingly daunting task for individuals across various backgrounds and qualifications. The process of finding a suitable job can be overwhelming and time-consuming, requiring a comprehensive understanding of the job market dynamics and effective strategies for job hunting. This project aims to develop a job search platform called "NiceJob" to facilitate the job search process for both job seekers and employers. The platform will provide a user-friendly interface with advanced search capabilities, allowing users to easily find relevant job opportunities. Job seekers will be able to create personalized profiles, upload their resumes, and apply for jobs through the platform. Employers, on the other hand, will have access to a comprehensive dashboard where they can post job listings, review applications, and manage the hiring process efficiently. The project will employ agile development methodologies to ensure flexibility and adaptability to changing requirements. The platform will be built using modern web technologies, including front-end frameworks for a responsive and intuitive user interface, and a scalable back-end architecture to handle the growing user base. Overall, "NiceJob" aims to revolutionize the job search experience by connecting job seekers with suitable employment opportunities and streamlining the recruitment process for employers.



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LIST OF ABBREVIATIONS

DFD	Dataflow Diagram
ERD	Entity Relationship Diagram
FK	Foreign Key
PK	Primary Key
SDLC	Software Development Life Cycle
UML	Unified Modelling Language
CV	Curriculum Vitae
CSS	Cascading Style Sheets
HTML	Hypertext Markup Language

CHAPTER ONE: INTRODUCTION

1.1 Introduction

In today's rapidly evolving job market, the employment search has become an increasingly daunting task for individuals across various backgrounds and qualifications. The process of finding a suitable job can be overwhelming and time-consuming, requiring a comprehensive understanding of the job market dynamics and effective strategies for job hunting [1]. Job seekers face numerous obstacles in their quest for suitable opportunities, such as fierce competition, limited job availability, and the need to stand out among other applicants. The emphasizes the magnitude of the challenges encountered by job seekers and sets the stage for further discussion. The acquisition of the right skills and attributes for effective job hunting is of paramount importance. While job search skills can be obtained from a variety of sources, including educational institutions, career counseling services, and personal experiences, they demand dedicated time and deliberate practice to be mastered. The competitive nature of the job market poses significant hurdles for individuals with advanced qualifications, leading to unemployment and underemployment. This underscores the need for solutions that address the specific challenges faced by highly educated job seekers [2]. This project acts as an intermediary between job seekers and companies, facilitating the process of connecting individuals to suitable job opportunities. The platform's primary objective is to receive job applications from those seeking employment and establish communication with companies to provide them with potential candidates. Additionally, the platform collaborates with companies to fulfill their immediate staffing needs by offering suitable candidates who have applied through the website. The website and its role in bridging the gap between job seekers and companies.

1.2 Problem Statement

The "NICEJOB" website serves as a platform for individuals seeking employment ("job seekers") to submit job applications. However, there is a lack of employment opportunities specifically tailored for beginners and individuals without specific certifications. Many job openings in the market require extensive qualifications and experience, leaving entry-level candidates with limited options. Additionally, other job search platforms often overlook part-time positions or roles related to events like Riyadh Season and other similar activities in Yemen. This creates a gap in the job market, hindering the opportunities available to beginners.

The "NICEJOB" project aims to address these problems by providing a user-friendly website that simplifies the job application process for individuals, offers guidance on effective job application strategies, expands access to a wide range of job opportunities, including part-time positions, and facilitates the evaluation and comparison of applicants based on relevant criteria. The project will focus on providing opportunities specifically tailored to beginners and individuals without specific certifications, allowing them to gain valuable work experience and contribute to the Yemen job market.

1.3 Objectives

The proposed project has several objectives aimed at addressing the challenges faced by job seekers and improving the efficiency of the recruitment process. These objectives include:

1. **Simplified User Experience:** To provide a clean, intuitive, and user-friendly interface that enhances the job search experience.
2. **Efficient Matching:** To implement advanced algorithms that accurately match job seekers with relevant job postings based on their skills, experience, and preferences.
3. **Personalization:** To offer personalized job recommendations and alerts based on user behavior and preferences.
4. **Employer Tools:** To equip employers with effective tools for posting jobs, filtering candidates, and managing applications.
5. **Connect job seekers with companies:** Our primary objective is to establish effective communication channels with companies seeking employees. We strive to match companies with suitable candidates from our pool of job seekers, enabling them to efficiently fulfill their hiring needs.
6. **Enhanced Search Functionality:** To provide robust and flexible search options, including keyword search, location filters, and job type categories, to help users find the most relevant job opportunities quickly.
7. **Mobile Accessibility:** To ensure that the job search engine is fully optimized for mobile devices, allowing users to search for jobs and manage their applications seamlessly from their smartphones and tablets.
8. **Data Security and Privacy:** To implement strong security measures to protect user data and ensure compliance with privacy regulations, giving users confidence in the safety of their personal and professional information.

By achieving these objectives, "NICEJOB" aims to create a thriving website that successfully connects job seekers with employment opportunities while meeting the recruitment needs of companies efficiently.

1.4 Scope

The scope of the "NICEJOB" project is focused on job seekers who are seeking employment specifically in Yemen. The project aims to provide a comprehensive platform tailored to the needs of job seekers and companies within the Yemen job market.

The scope of the project for job seekers includes, but is not limited to, the following areas:

1. **Job Seekers:**

- Registration and profile creation: The project will allow job seekers to register on the platform and create detailed profiles highlighting their qualifications, experiences, skills, and preferences specifically for the Yemen job market.
- CV optimization: The project will provide guidance and assistance to job seekers in optimizing their CVs to increase their chances of being noticed by employers in Yemen.
- Networking and career development: The project will include resources, tips, and guidance specifically tailored to help job seekers in Yemen improve their networking skills, enhance their career development, and effectively navigate the Yemen job market.
- Part-time opportunities: The project will feature part-time job opportunities that do not require specific certifications, giving job seekers the chance to gain work experience and supplement their income.

The scope of the project for companies in Yemen includes, but is not limited to, the following areas:

2. Companies:

- Post Job Vacancies: The project will include a feature that allows companies in Yemen to post job vacancies specific to their company and the Yemen job market, including part-time positions.
- Scan Job Seekers List: The project will enable companies to view and review the list of job seekers, including their names, areas of expertise, educational background, and CVs, specifically focusing on job seekers interested in opportunities in Yemen.

The "NICEJOB" project is focused on job seekers who are seeking employment specifically in Yemen. The project aims to provide a comprehensive platform tailored to the needs of job seekers and companies within the Yemen job market.

1.5 Significance of the Problem

The significance of the problem lies in addressing the challenges and issues faced by both job seekers and companies in the recruitment process. Here are the key reasons why addressing these problems is important:

1. Improved job seeker outcomes: By enhancing the quality of CVs and providing personalized feedback, job seekers have a higher chance of presenting their qualifications and experiences effectively. This leads to improved visibility to potential Job seekers and increases their chances of securing employment. When job seekers have well-structured and comprehensive CVs, they can better showcase their skills, experiences, and qualifications, making a strong impression on Job seekers. [5]
2. Increased company satisfaction: A well-functioning recruitment platform that efficiently connects companies with suitable candidates leads to increased company satisfaction. When companies can

find the right talent quickly and easily, it improves their hiring process and increases their confidence in the platform. This positive experience encourages more companies to utilize the platform's services and collaborate on future recruitment needs. It builds trust and credibility in the platform, leading to long-term partnerships and a growing network of companies. [6]

3. **Economic growth and development:** A robust job market is essential for economic growth and development. By facilitating employment opportunities and connecting job seekers with suitable positions, the platform contributes to economic growth. It reduces unemployment rates, improves job market efficiency, and supports the growth of businesses by providing them with a talented and qualified workforce. When individuals find suitable employment, they contribute to the economy through their productivity, spending power, and overall economic participation. [2]

Overall, addressing the identified problems brings significant benefits to both job seekers and companies. It leads to improved job seeker outcomes, increased employment opportunities, enhanced efficiency in recruitment, better matching of skills and qualifications, higher company satisfaction, and overall economic growth and development. By creating a platform that effectively connects job seekers and companies, the platform plays a crucial role in shaping the labor market and fostering a thriving economy.

1.6 Limitations of the Project

Every project, regardless of its nature, is bound to face certain limitations that can impact its scope, effectiveness, and overall success. Understanding and addressing these limitations is crucial for project managers and stakeholders to mitigate risks and maximize project outcomes. In the case of this project, several potential limitations should be taken into consideration. This section will provide a comprehensive and detailed overview of the project's limitations, highlighting the key areas where constraints may arise.

1. **Limited reach and accessibility:** The project's impact may be limited by factors such as internet access and technological infrastructure. Job seekers and companies with limited access to the internet or technology may face difficulties in utilizing the platform effectively. This could result in a narrower reach and reduced opportunities for those who do not have sufficient access to participate in the project.
2. **Reliance on user-submitted information:** The project relies on job seekers to provide accurate and up-to-date information about their qualifications, experiences, and skills. However, there is a possibility of inaccurate or misleading information being presented, which can impact the matching process and the overall effectiveness of the platform. Implementing rigorous verification processes can help mitigate this limitation but may introduce additional complexities and time constraints.

3. **Dependence on employer participation:** The success of the project depends on the active participation and engagement of companies seeking employees. If there is limited employer engagement or a lack of available job opportunities, it can restrict the platform's ability to connect job seekers with suitable employment options. Building and maintaining strong partnerships with companies is crucial to ensure a consistent flow of job opportunities.
4. **Job market fluctuations:** The project may be affected by fluctuations in the job market, economic conditions, and industry-specific trends. Changes in the availability of job opportunities, shifts in demand for specific skills, or economic downturns can impact the project's ability to provide employment opportunities and successful placements. Adapting to changing market dynamics and diversifying the range of available job opportunities can help mitigate this limitation.
5. **Privacy and data security concerns:** The project involves collecting and storing personal information from job seekers and companies. Ensuring the privacy and security of this data is of utmost importance to maintain user trust.

While these limitations exist, they can be addressed through strategic planning, partnerships, user feedback integration, and continuous improvement efforts. By acknowledging and actively working to overcome these limitations, the project can maximize its impact and provide meaningful support to job seekers and companies in the recruitment process.

1.7 Advantages of the Project

The "NICEJOB" project offers several significant advantages that benefit both job seekers and companies. These advantages contribute to a more efficient and effective job market, improve the overall experience of job seekers, and facilitate better connections between job seekers and potential Job seekers. Let's explore these advantages in detail:

1. Diverse Range of Job Opportunities:

The platform provides job seekers with access to a diverse range of job opportunities across various industries and job roles. This broadens the scope of available positions, increasing the chances of job seekers finding suitable employment. Additionally, it allows job seekers to explore different industries and roles, facilitating career growth and expansion.

2. Personalized Guidance and Recommendations:

Job seekers receive personalized guidance, feedback, and recommendations throughout their job search journey. This includes CV optimization suggestions, training and skill development opportunities, and networking and career development resources. The personalized nature of these recommendations helps job seekers tailor their approach and improve their chances of success in the job market.

3. Increased hiring opportunities:

The platform increases the visibility of job seekers by showcasing their profiles and qualifications to potential Job seekers. This exposure maximizes their chances of being noticed by recruiters and increases the likelihood of receiving interview invitations and job offers. By enhancing job seekers' visibility, "NICEJOB" improves their opportunities for employment Platform:

The "NICEJOB" project provides a user-friendly platform that is accessible to job seekers of all backgrounds and technical abilities. The interface is designed to be intuitive and easy to navigate, ensuring a positive user experience. Job seekers can easily create and update their profiles, search for job opportunities, and interact with the platform's features without facing unnecessary complications.

4. Continuous Support and Updates:

The project offers continuous support to job seekers, ensuring that they receive ongoing assistance throughout their job search process. This includes updates on new job opportunities, training programs, and industry trends. By providing consistent support and updates, "NICEJOB" keeps job seekers informed and engaged, enhancing their overall experience and keeping them motivated in their job search.

1.8 Proposed System

The proposed website is an expansive and user-friendly online platform designed to revolutionize the job market and address the challenges faced by job seekers. This comprehensive website aims to provide job seekers with an innovative and efficient solution to enhance their job search process and increase their chances of finding suitable employment. The primary objective of the proposed website is to create a centralized and user-centric platform that caters to the needs of job seekers. The platform aims to provide a seamless and effective experience for job seekers by offering a range of features and functionalities that streamline the job search process, enhance their qualifications and skills, and connect them with relevant job opportunities. [2]

1.8.1 User Registration and Profile Creation:

One of the key features of the "NICEJOB" website is the user registration and profile creation system. Job seekers can easily create personalized accounts by providing their essential personal information, such as name, contact details, and email address. Once registered, job seekers can proceed to build detailed and comprehensive profiles that highlight their qualifications, experiences, skills, and preferences.

1.8.2 Advanced Job Search and Filtering:

The "NICEJOB" website incorporates advanced job search and filtering capabilities to facilitate a refined and targeted job search process. Job seekers can utilize powerful search filters to narrow down their job search based on specific criteria such as location, industry, and job title.

1.8.3 CV Optimization Tools:

To assist job seekers in presenting their qualifications effectively, "NICEJOB" offers advanced CV optimization tools. Job seekers can upload their CVs and analyze the CVs manually.

1.8.4 Company Profiles and Job Postings:

The website features comprehensive company profiles that provide job seekers with valuable insights into potential Job seekers. Job seekers can access information about the company's background, culture, values, and available job openings. The platform also allows companies to post detailed job vacancies, specifying the required qualifications, skills, and responsibilities for each position. [2]

1.8.5 Networking and Community Building:

"NICEJOB" fosters networking and community building by providing job seekers with opportunities to connect with industry professionals, seek guidance, and expand their professional network. This feature aims to enhance job seekers' networking skills, career development, and overall employability.

1.9 System Architecture

System architecture refers to the design of a structure's components or elements [4]. In the case of the proposed system, it will be implemented as a PHP website with a MYSQL Database. Such systems often adopt a three-tier architecture, as depicted in Figure 1.2. A three-tier architecture is a client-server model where the functional logic, data access, data storage, and user interface are developed and managed as separate modules on different platforms.

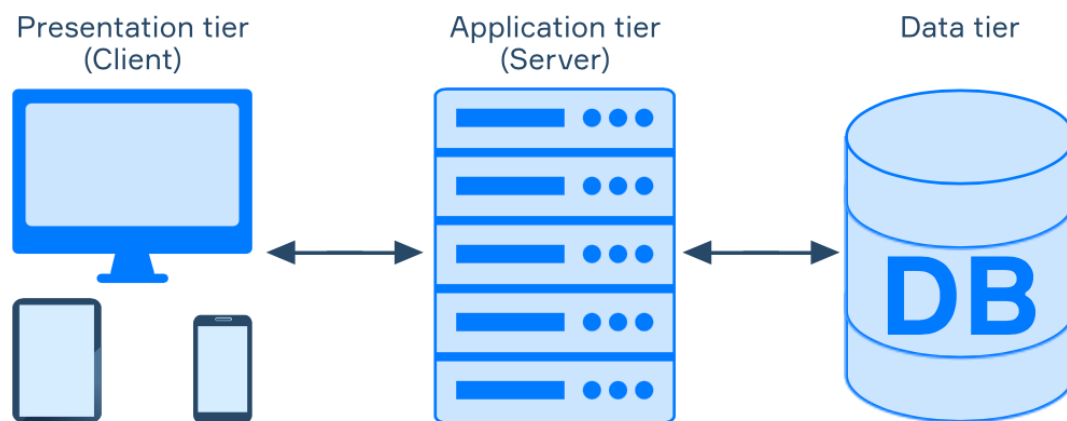


Figure 1.1. System Architecture

1.10 Conclusion

In conclusion, this chapter has shed light on the numerous challenges that job seekers encounter during their job search, ranging from difficulties in finding suitable positions to navigating through the vast array of available opportunities. These challenges can often lead to frustration, time wastage, and

decreased productivity. However, the "NICEJOB" project has been introduced as a comprehensive solution aimed at enhancing job seekers' productivity and improving their overall job search experience.

The primary objective of the "NICEJOB" project is to streamline and simplify the job-seeking process. By providing a centralized platform, job seekers gain access to a wide range of employment opportunities from companies spanning various industries. This platform acts as a bridge, connecting job seekers with potential employers and facilitating the interaction between them.

To achieve this goal, the project offers a plethora of services and opportunities to job seekers. These include features such as advanced job search and filtering options, personalized job recommendations, CV optimization tools, and networking functionalities. By providing these resources, the project aims to empower job seekers with the tools and information they need to make informed decisions and improve their chances of securing suitable employment.

Throughout the chapter, the project's objectives and scope have been thoroughly discussed, highlighting the emphasis on facilitating the matching process between job seekers and suitable job opportunities. By focusing on this aspect, the project aims to address the gap between job seekers' qualifications and the requirements of employers.

Furthermore, the current scenario and limitations of the project have been carefully examined. It is essential to understand the existing landscape and potential challenges that may arise during implementation. By acknowledging these limitations, appropriate strategies and solutions can be developed to overcome them and ensure the successful execution of the project.

In terms of project management, a Gantt chart has been provided to outline the various tasks and their corresponding timelines. This chart serves as a visual guide, enabling effective monitoring of progress and ensuring that milestones are achieved promptly. By utilizing this project management tool, the project can proceed systematically and efficiently, minimizing delays and maximizing productivity.

Additionally, a system architecture based on a three-tier model has been proposed for the project. This architecture separates the components into distinct tiers, including the presentation tier, application tier, and data tier. By adopting this modular approach, the project ensures scalability, maintainability, and adaptability to future enhancements or changes. The separation of concerns allows for efficient development, testing, and maintenance of each tier, resulting in a robust and reliable system.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

In this chapter, we will embark on a comprehensive exploration of the existing literature related to online job search websites. The aim is to establish a strong foundation of knowledge and understanding regarding the concepts, theories, and practices that underpin the development and effectiveness of such platforms. By examining the existing body of research, we can identify key insights, best practices, and potential gaps in knowledge that will inform the development process of "NiceJob."

2.2 Job Portals in Yemen

Job portals in Yemen serve as online platforms connecting job seekers and employers. They provide extensive databases of job listings across various industries. Job seekers can create profiles, upload CVs, and apply for jobs, while employers can post vacancies, search for candidates, and manage the recruitment process. These portals offer features such as job alerts, and networking opportunities. Mobile applications enhance accessibility. Popular job portals in Yemen include Bayt.com, LinkedIn, and Mihnati.com. Utilizing these portals helps job seekers find suitable opportunities and allows employers to identify qualified candidates, contributing to the overall job search and recruitment process in the country [5].

2.3 Job Portals Objectives

We list the three main objectives of jobs as follows:

1. **Connecting Job Seekers and Employers:** Job portals serve as a platform to connect job seekers with employment opportunities by providing a centralized database of job listings and facilitating communication between candidates and employers.
2. **Enhancing Job Seeker Outcomes:** Job portals aim to improve job seeker outcomes by offering features such as job alerts, personalized recommendations, and career guidance to assist in the job search process and increase the chances of securing suitable employment.
3. **Simplifying the Recruitment Process for Employers:** Job portals streamline the recruitment process for employers by providing tools to post job vacancies, manage applications, and search for qualified candidates, saving time and resources while ensuring access to a diverse talent pool.

2.4 Project Feasibility

Feasibility is a study designed to reveal whether a project or plan is feasible, it is an assessment of the feasibility of a proposed project. The proposed system works to provide a user-friendly platform for job seekers, offering features such as user registration and profile creation, advanced job search and filtering, CV optimization tools, comprehensive company profiles, and networking opportunities. Job seekers can create personalized accounts and build detailed profiles highlighting their qualifications and preferences. The advanced search and filtering capabilities enable a refined job search based on specific

criteria. The CV optimization tools assist job seekers in presenting their qualifications effectively. Comprehensive company profiles offer valuable insights into potential employers. Additionally, the networking feature allows job seekers to connect with professionals and expand their professional network, enhancing their overall employability.

2.4.1 Technical Feasibility

The assessment of technical feasibility involves examining all the technical aspects associated with the project. This includes evaluating the project's scale, technical specifications, and the necessary technology for implementation [6]. The planned system will be created as a website using the PHP programming language and a MySQL database server. Consequently, the proposed system is considered technically feasible since obtaining the required resources to develop it should not pose significant challenges.

2.4.2 Operational Feasibility

The operational feasibility study is the study that studies the capabilities of the organization undertaking the project and determines their ability to meet the standards of the work environment, this system will be an online website, so all job seekers can use it from any laptop or mobile device that is connected to the internet, so the proposed project have high operational feasibility.

2.4.3 Economic Feasibility

Developing a website such as our project does not cost a lot of money and running it does not require a high operational cost, all the resources needed for the developing website are free software and available to download, so the proposed project carries a high economic feasibility [6].

Table 2-1. Financial feasibility

Requirements	Type	Quantity	Cost	Total
Hardware	Laptop	4	3375	13500 SAR
Software	Win10	4	Free	Free
	Laravel 10.x	4	Free	Free
	MySQL 8.0	4	Free	Free
	Nginx 1.20	1	Free	Free
	Notepad	4	Free	Free
	AnyDesk	4	Free	Free
	Docker	4	Free	Free
	Project Libre	4	Free	Free
	Adobe Illustrator	4	Free	Free
	MS word	4	Free	Free
Others	Domain Name	3	319/1Year	955 SAR
	VPS Hosting	36	150/month	5400 SAR
	Internet	700GB	Unlimited	2857 SAR
Total				22712 SAR

2.4.4 Schedule Feasibility

As explained in the project plan in Chapter One figure 1-1, it is possible to easily complete the project within the specified time. A Gantt chart is a tool that is used in project management to represent a project plan over time, a Gantt chart includes the start and end dates of tasks, and milestones [3].

2.4.5 Project Work Breakdown

Work breakdown structure (WBS) is a method used to complete complex projects that involve multiple steps. It involves breaking down the work into smaller segments, allowing team members to work on tasks simultaneously. This approach enhances team productivity and streamlines project management by dividing the workload and facilitating efficient coordination among team members [7].

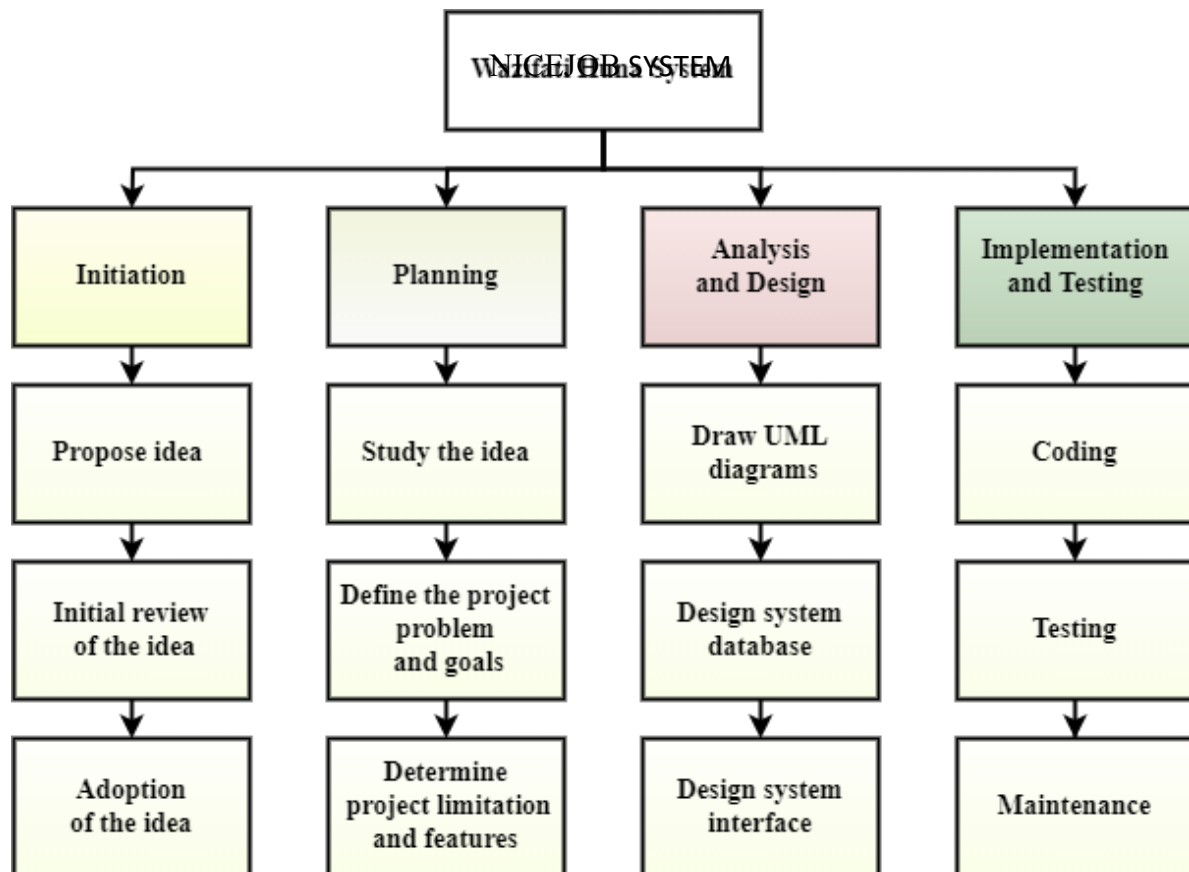


Figure 2.1. Project Work breakdown

2.5 Related Work

In this section, some websites like the proposed system will be studied and compared.

2.5.1 Glassdoor

Glassdoor is a popular job search website that provides valuable company reviews, salary information, and job listings. It offers job search tools and resources, making it a go-to platform for job seekers. Glassdoor also provides recruitment solutions for companies. However, it lacks certain features that are offered by the "NICEJOB" job portal. For instance, Glassdoor does not provide review CV services, companies cannot view job applicants and select one or more candidates, and it does not offer part-time job opportunities. By offering these additional features, the "NICEJOB" job portal aims to address the limitations of Glassdoor and provide a more comprehensive platform for job seekers and companies[8].



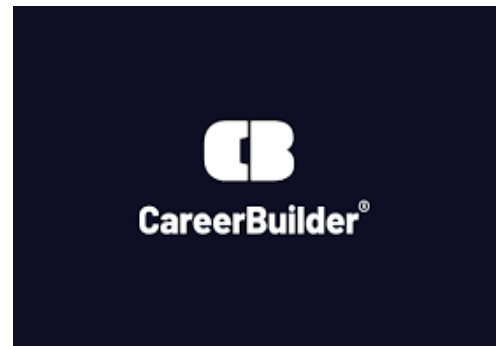
2.5.2 Indeed

Indeed is another well-known job search website that aggregates job listings from various sources and offers job search tools and resources. It provides recruitment solutions for companies, making it a valuable platform in the job market. However, similar to Glassdoor, Indeed lacks certain features offered by the "NICEJOB" job portal. For example, it does not provide review CV services, companies cannot view job applicants and select candidates, and it does not offer part-time job opportunities. By incorporating these additional features, the "NICEJOB" job portal aims to differentiate itself and provide a more comprehensive solution for job seekers and companies [9].



2.5.3 CareerBuilder

CareerBuilder is a well-established job search website that offers job listings, career advice, and resources for job seekers. It also provides recruitment solutions for companies, making it a valuable platform in the job market. However, similar to the previous platforms, CareerBuilder lacks certain features provided by the "NICEJOB" job portal. It does not offer review CV services, companies cannot view job applicants and select candidates, and it does not provide part-time job opportunities. The "NICEJOB" job portal aims to address these limitations by offering comprehensive features such as CV optimization, training opportunities, and connecting job seekers with companies[10].



2.5.4 Jobvite

Jobvite is a recruitment software that offers applicant tracking, social recruiting, and employee referral tools. It provides recruitment solutions for companies and offers job search tools and resources. While Jobvite is a valuable platform, it does not provide review CV services, companies cannot view job applicants and select candidates, and it does not offer part-time job opportunities. These are features that set the "NICEJOB" job portal apart. By offering these additional features, the "NICEJOB" job portal aims to provide a more comprehensive platform for job seekers and companies, addressing the limitations of existing platforms[11].



2.5.5 *SmartRecruiters*

SmartRecruiters is a platform that offers applicant tracking, recruitment marketing, and candidate engagement tools. It provides recruitment solutions for companies and offers job search tools and resources. However, similar to the other platforms discussed, SmartRecruiters lacks certain features provided by the "NICEJOB" job portal. It does not offer review CV services, companies cannot view job applicants

and select candidates, and it does not provide part-time job opportunities. By incorporating these additional features, the "NICEJOB" job portal aims to provide a more comprehensive solution for job seekers and companies, bridging the gaps in existing platforms[12].



2.5.6 *Comparative Study*

In this comparative study, we will examine the key features and advantages of the “NICEJOB” job portal in comparison to the previously mentioned related work platforms: Glassdoor, Indeed, CareerBuilder, Jobvite, and SmartRecruiters.

1. Diverse Range of Job Opportunities:

“NICEJOB”: The platform offers a diverse range of job opportunities across various industries and job roles, allowing job seekers to explore different options and expand their career prospects.

Related Work: Glassdoor, Indeed, CareerBuilder, Jobvite, and SmartRecruiters also provide job listings from various sources, offering a broad range of job opportunities.

2. Personalized Guidance and Recommendations:

“NICEJOB”: The platform provides personalized guidance, feedback, and recommendations to job seekers, including a unique feature of reviewing CVs and suggesting improvements to applicants.

Related Work: Glassdoor, Indeed, CareerBuilder, Jobvite, and SmartRecruiters primarily focus on job listings and recruitment solutions, providing limited personalized guidance and recommendations.

3. Increased Hiring Opportunities:

"NICEJOB": The platform enhances the visibility of job seekers, increasing their chances of being noticed by recruiters and receiving job offers. Additionally, it offers part-time job opportunities to individuals in Yemen without the requirement of specific qualifications, only basic skills such as language proficiency.

Related Work: Glassdoor, Indeed, CareerBuilder, Jobvite, and SmartRecruiters also aim to connect job seekers with companies, but they may not offer the same level of visibility, CV review service, and part-time job opportunities as “NICEJOB.”

4. Accessible and User-Friendly Platform: “NICEJOB”: The platform provides a user-friendly interface accessible to job seekers of all backgrounds and technical abilities, ensuring a positive user experience.

Related Work: Glassdoor, Indeed, CareerBuilder, Jobvite, and SmartRecruiters also prioritize user experience, but the ease of use and interface design may vary.

4. Continuous Support and Updates:

“NICEJOB”: The platform offers continuous support and updates to job seekers, keeping them informed about new opportunities, training programs, and industry trends.

Related Work: Glassdoor, Indeed, CareerBuilder, Jobvite, and SmartRecruiters may provide updates on job listings and industry trends, but they may not offer the same level of ongoing support and engagement, including CV review and part-time job opportunities, as “ NICEJOB.”

Overall, the "NICEJOB" job portal distinguishes itself by offering a comprehensive set of features that address the needs of job seekers throughout the job search process. It provides a diverse range of job opportunities, personalized guidance with a unique CV review service, increased hiring opportunities including part-time jobs, an accessible platform, and continuous support and updates. While the related work platforms offer job listings and recruitment solutions, they may not provide the same level of personalized guidance, visibility, CV review service, and part-time job opportunities as “NICEJOB.”

2.6 Conclusion

In this chapter, we discussed a literature review of relevant works related to the proposed projects. A comparison was made between previous projects and the proposed system, highlighting their respective advantages and disadvantages. The findings indicate that our site offers comprehensive services that cater to the job seeker's needs throughout the job search process. Furthermore, the feasibility of the project was examined, considering technical feasibility, economic feasibility, operational feasibility, and project work breakdown structure.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

The system analysis stage is a crucial part of the software project life cycle and marks the actual initiation of the software development process. Software systems analysis is a dynamic field that constantly evolves with new techniques and approaches to accurately capture, comprehend, and develop software systems that are more efficient and effective. The primary goal of the analysis phase is to determine the desired outcomes expected by the user for the proposed system. During this phase, the overall business requirements of the system will be translated into specific functional requirements, while also identifying nonfunctional requirements.

3.2 Systems Development Life Cycle (SDLC)

The Systems Development Life Cycle (SDLC) encompasses multiple stages that a project undergoes before it reaches a usable state. These stages include planning, requirements gathering, analysis, design, implementation, and testing. The SDLC provides a structured approach for developers to design, develop, and deliver software of high quality. It outlines the necessary tasks that must be executed throughout the process to ensure successful software development [10].

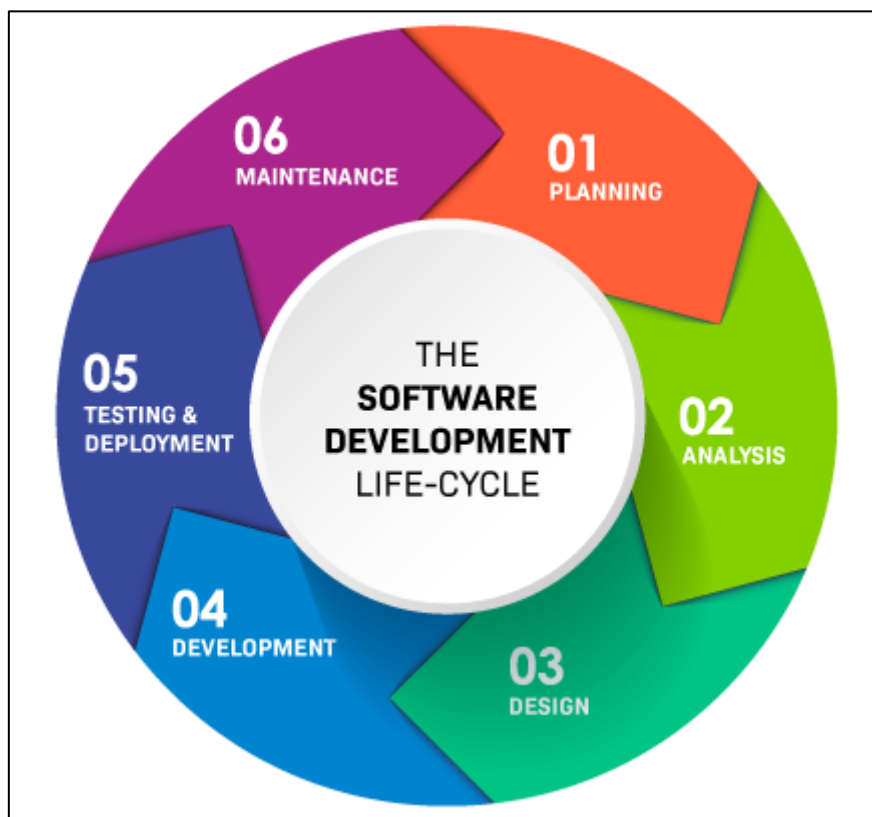


Figure 3.1. SDLC

3.2.1 Phases of SDLC

1. The planning phase serves as the initial stage of a project, where an overall view of the project is established. It involves determining the project's scope, and goals, and identifying the

business requirements. Activities in this phase include identifying system users and stakeholders [11].

2. The analysis phase follows a structured process and involves translating the business requirements of the proposed system from a high-level perspective to specific functional requirements. Non-functional requirements are also identified during this phase. UML diagrams may be created to represent the proposed system [11].
3. The design phase is a crucial step that bridges the gap between theoretical aspects, such as planning and analysis, and the practical implementation of the system. It focuses on creating a solution that aligns with the requirements. This phase marks the beginning of transforming requirements into a system solution to fulfill the identified needs [11].
4. The implementation phase begins after the system has been tested and accepted by the user. It involves installing the system to support the intended business functions. System performance is compared against the performance objectives established during the planning phase [12].
5. The testing and maintenance phase aims to ensure that the developed system meets the required business requirements and achieves the project objectives. Extensive testing is conducted to verify the functionality of all system buttons, menus, and functions. This phase ensures that the system operates as expected [11].

3.3 Agile Methodology

The system development methodology serves as a framework for organizing, planning, and controlling the project development process. It defines how the phases of the SDLC are implemented. There are various software methodologies available, including Agile, Agile, Extreme Programming, Scrum, and Spiral, each with its strengths and drawbacks. For this particular project, When the project requirements are not well-defined initially or are subject to change, Agile methodology becomes more suitable than other. Agile is characterized by flexibility and the ability to adapt to changes during the development cycle. Agile relies on iterative development processes and gradual delivery of features. Instead of waiting until all requirements are complete before starting implementation, the project is divided into short iterations during which specific features are developed and delivered.

3.3.1 Agile Phases

1. Requirements Gathering: Identifying and analysing user needs to define the project scope.
 - Ensures the product will meet customer expectations.
2. Design: Developing the detailed design of the product based on the identified requirements.
 - Transforms the design into working software.

3. Development: Writing and implementing the code to execute the defined requirements.
 - Transforms the design into working software.
4. Testing: Verifying that the software functions according to the specified requirements.
 - Ensures the quality of the final product.
5. Deployment: Releasing the developed software into the production environment.
 - Makes the product available to end-users.
6. Review: Reviewing the project outcomes and identifying opportunities for improvement.
 - Enhances the development process for future projects.

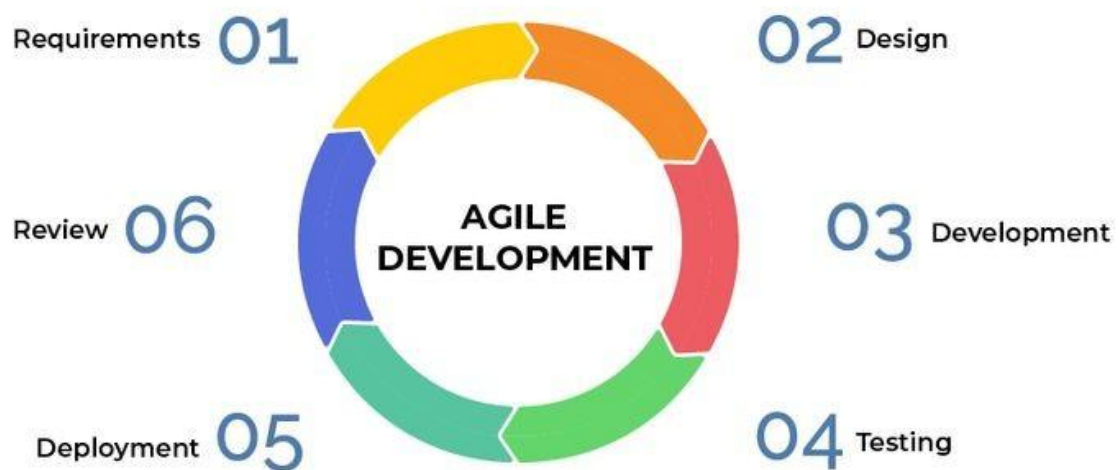


Figure 3.2. Agile phases

3.3.2 Advantages of the Agile Model

The Agile model offers several advantages, especially in dynamic and fast-paced environments where flexibility and customer satisfaction are key. Here are some of the main advantages:

1. Adaptability and Flexibility:
 - The Agile model is characterized by flexibility and the ability to adapt to changing requirements and evolving needs during the project lifecycle.
 - It allows for iterative improvements and modifications to the product.
2. Early Delivery of Value:
 - Agile focuses on the incremental and iterative delivery of value to the client, rather than waiting until the entire project is complete.
 - This provides early feedback and an opportunity to improve the product.
3. Focus on Core Needs:
 - Agile places a greater emphasis on the client's core needs and priority challenges.
 - This helps to reduce unnecessary features.
4. Client Involvement:
 - The Agile model encourages continuous interaction with the client and end-users.
 - This leads to a product that is more aligned with their needs.
5. Cohesive and Effective Teams:
 - Agile relies on small, cross-functional, and self-organizing teams.
 - This fosters collaboration, commitment, and productivity.

These advantages make Agile a highly effective approach in environments where requirements may evolve, and speed, quality, and customer satisfaction are top priorities.

3.4 System Requirements

System requirements encompass all the necessary specifications and guidelines provided to the user to facilitate the implementation of a project. In our specific case, the following requirements need to be available for the development of our project.

3.4.1 Technical Requirements

The hardware and software prerequisites necessary for the successful implementation of the proposed system are presented in the following tables.

Table 3-1. Hardware Requirements

Device	Specifications	Details
Personal Computer or Laptop	CPU: Core i5	A computer with a Core i5 processor for installing the necessary software and developing the website.
	RAM: 8GB	Sufficient memory capacity (8GB) to handle the software requirements and facilitate smooth operation.
Internet	Any type of internet connection	An internet connection is required for downloading necessary software and conducting online searches.

Table 3-2. Software requirements

Software	Specifications	Details
Windows 10 or 11	Operating System	The personal computer or laptop should have Windows 10 or 11 installed as the operating system.
Visual Studio Code IDE	Latest version	The Visual Studio Code IDE, in its latest version, will be used to create website interfaces and write PHP code.
MySQL	Version 8.0	MySQL version 8.0 will be utilized to create the system's database.

3.4.2 Project Development Tools

Table 3-3. Development requirements

Item	Specifications	Details
PHP	Version 7.4	The main programming language used for website development.
HTML	Version 5.0	To create user interfaces and structure web page content.
CSS	Version 3	Used for styling web pages, including layout, colors, and typography.

JavaScript	-	For writing client-side scripts to enhance website functionality.
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3.4.3 Functional Requirements

Functional requirements outline the desired behavior actions that the system should exhibit. Each participant involved in the proposed system referred to as an actor, possesses distinct functional requirements. The functional requirements for each actor are as follows [14].

1. **User Registration:** Users should be able to create an account on the website by providing their personal details such as name, email address, and password.
2. **Job Search:** Users should be able to search for job listings based on various criteria such as job title, location, industry, and keywords.
3. **Resume Builder:** The website should offer a resume-building feature that allows users to create and edit their resumes. Users should be able to save and download their resumes.
4. **Company Profiles:** The website should provide detailed information about companies that are hiring, including their mission, values, culture, and employee reviews.
5. **User Profile Management:** Users should be able to manage their profiles, including updating their personal information, and uploading and managing their resumes.

Functional user requirements:

- ✓ The system must provide a login interface for the system, and login must be according to the authority given to the user. (Employer or job seeker).
- ✓ The system provides the user with the ability to browse and view advertisements, offers, and articles for opportunities related to employment.
- ✓ The system (employers) can control the job offer services to be advertised.
- ✓ The system sends and receives requests in general.
- ✓ The system stores the data of researchers on job opportunities and saves the CV in the database.
- ✓ That the user (job seeker) can edit, add or delete his CV and update it.
- ✓ That the user (employer) can modify, add, or delete the job that was announced.
- ✓ The employer should be able to search the CV stored in the database or the job should be advertised.

The system will enable the user to obtain the following services:

- Employers.
- Job seeker.
- Diverse jobs.
- Today's Jobs.
- Explore careers by publisher;

3.4.4 Non-functional Requirements

Non-functional requirements describe the properties of the system [14].

1. **Performance:** The website should be fast and responsive, providing quick search results and seamless navigation between pages. It should be able to handle a large number of concurrent users without significant slowdowns.
2. **Security:** The website should implement appropriate security measures to protect user data, including secure user authentication, encryption of sensitive information, and regular backups of data.
3. **Usability:** The website should have a user-friendly interface with intuitive navigation and clear instructions. It should be accessible to a wide range of users, including those with disabilities, by following accessibility guidelines.
4. **Compatibility:** The website should be compatible with different web browsers (e.g., Chrome, Firefox, Safari) and devices (desktops, laptops, tablets, mobile phones) to ensure a consistent user experience across platforms.
5. **Integration:** The website should be able to integrate with other systems or platforms, such as social media platforms for sharing job listings or third-party services for resume parsing and applicant tracking.
6. **Maintenance and Support:** The website should have a maintenance plan in place to address any bugs, issues, or security vulnerabilities that may arise. It should also provide support channels (e.g., email, chat, FAQs) for users to seek assistance when needed.

3.5 UML (Unified Modelling Language)

UML is a graphical modeling language that provides a formula to describe the key elements of software systems. This language is used to create diagrams to describe computer programs in terms of their components, elements, or processes of the program [13].

3.5.1 Use Case Diagram

Use case diagrams are used to collect system requirements, including those relating to internal and external factors, to capture the dynamic element of a system [13].

a website that connects job seekers with potential employers, allowing them to search for and apply to job openings. The Search for Jobs use case allows job seekers to search for job openings based on their preferred location, job title, and industry. This use case involves interactions between the user interface and the job portal's job search engine. Job seekers can search for job openings easily and find relevant job opportunities that match their skills and experience.

3.5.2 Sequence Diagram

A sequence diagram is a visual representation that illustrates the interaction and collaboration between objects. It serves to define the sequence of events required for a desired outcome. Also known as a timing diagram, event diagram, or event scenario, a sequence diagram plays a vital role in specifying and documenting the flow of activities between objects [14].

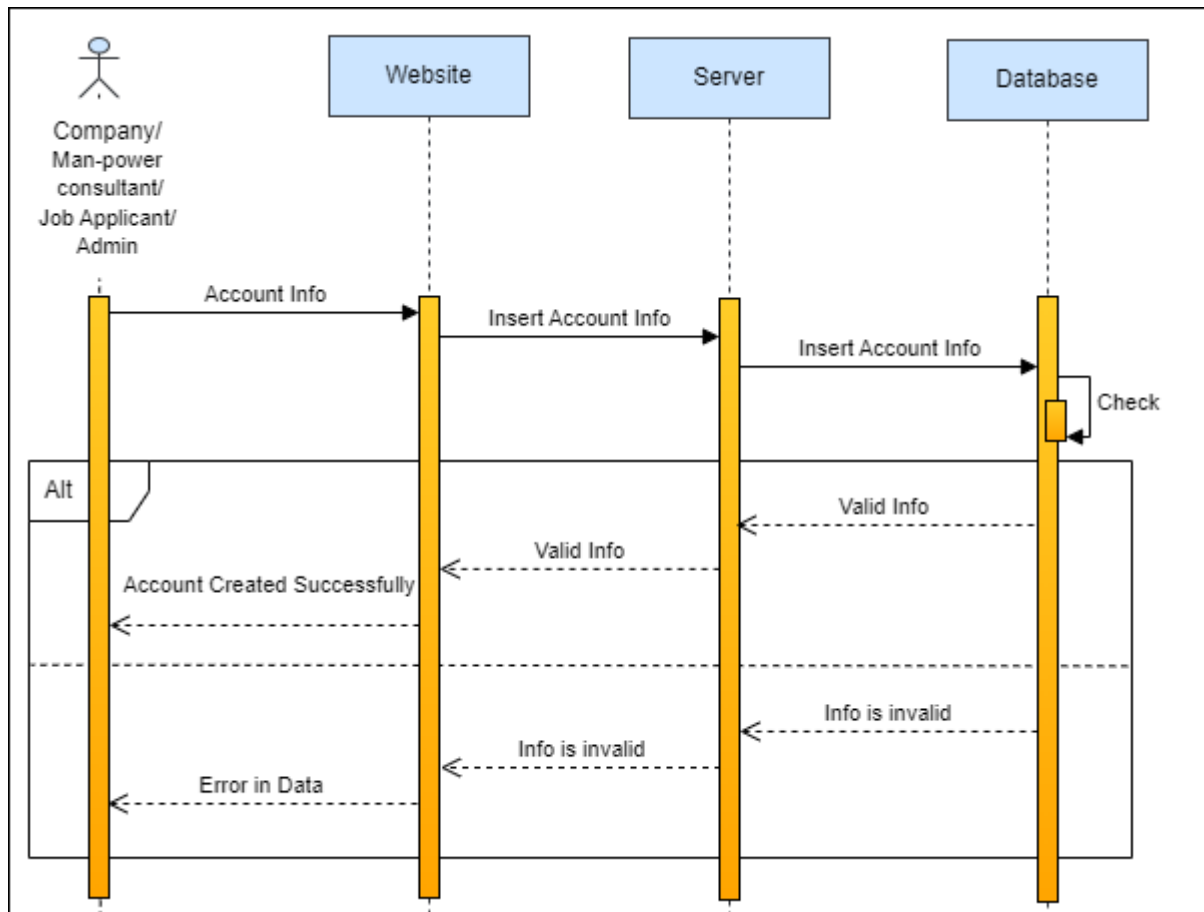


Figure 3.7. Register sequence diagram

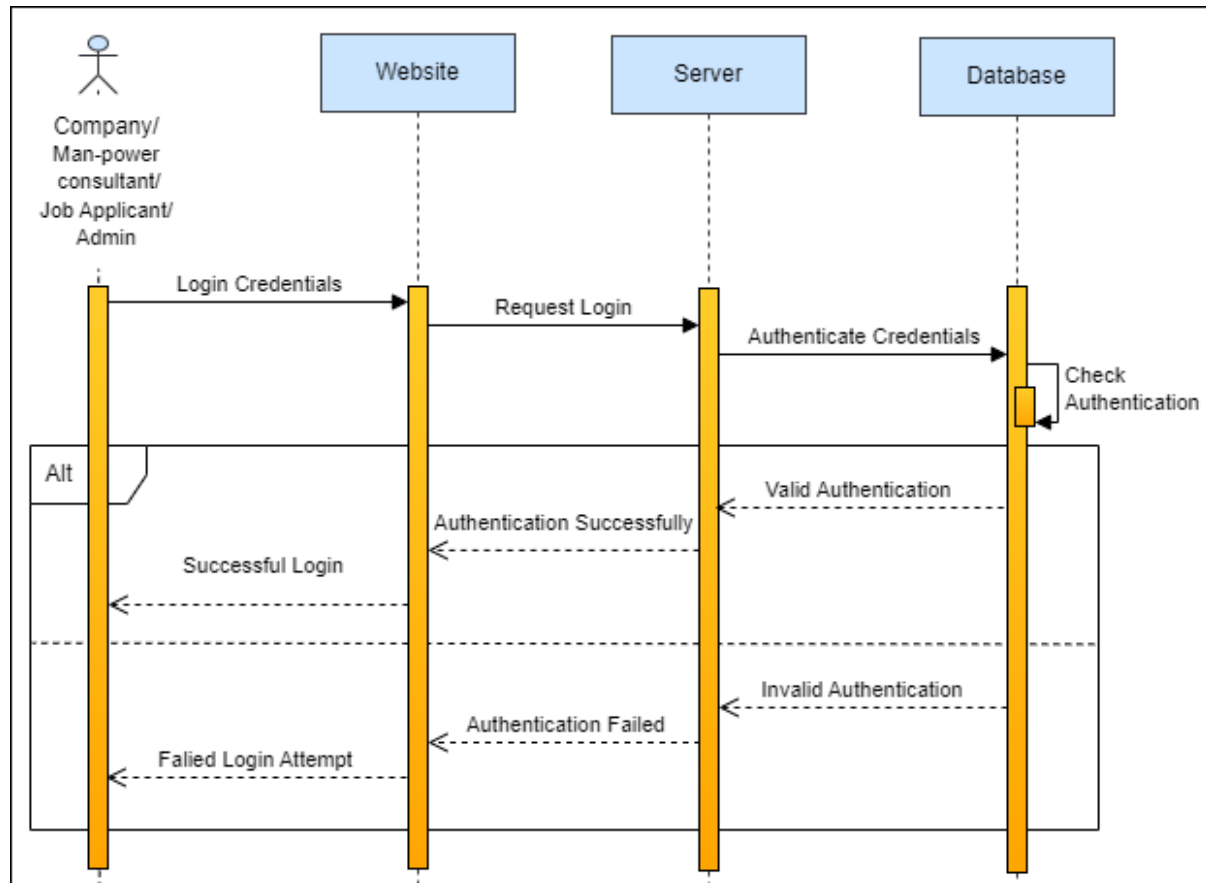


Figure 3.8. Login sequence diagram

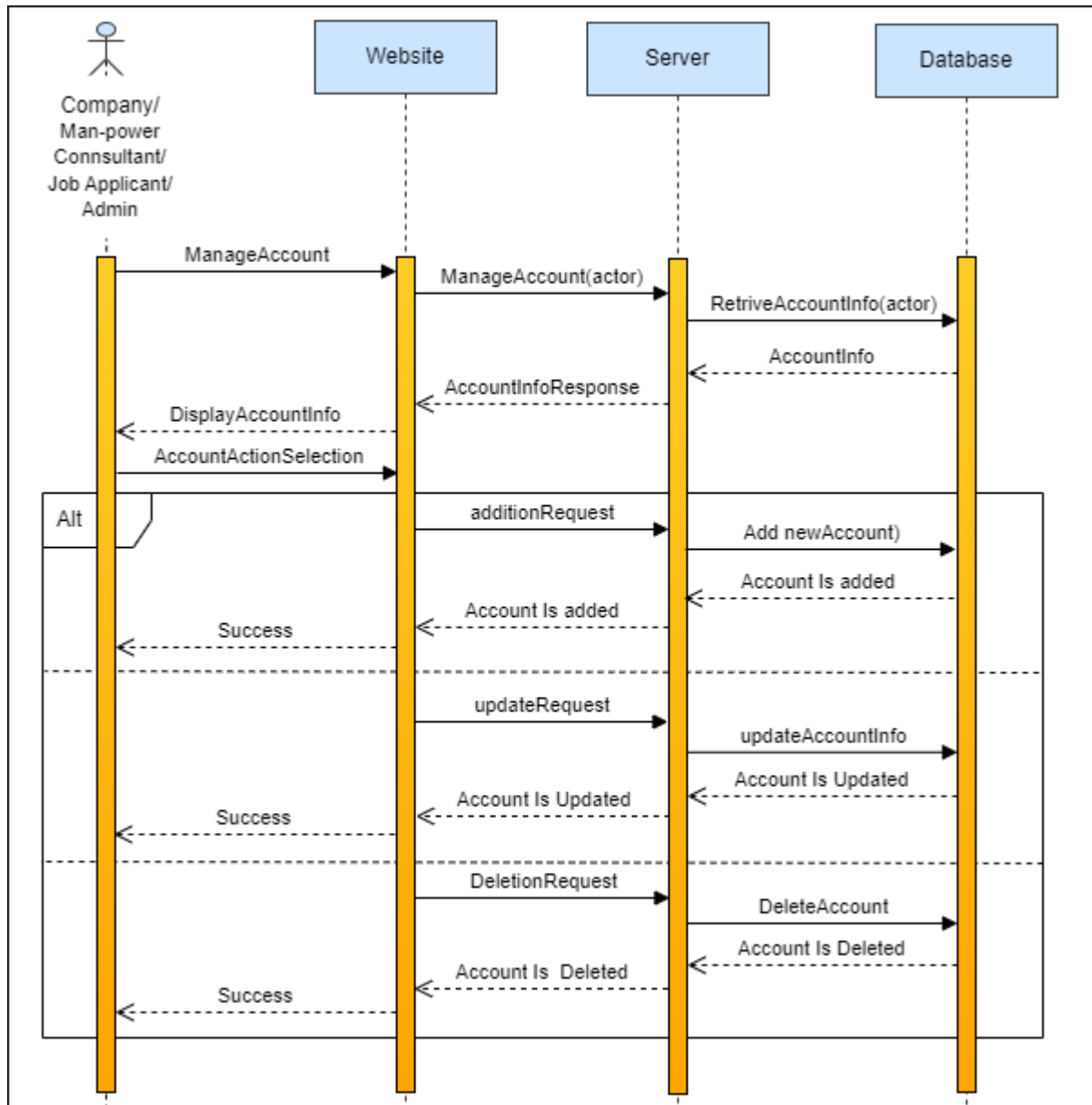


Figure 3.9. Manage Account sequence diagram

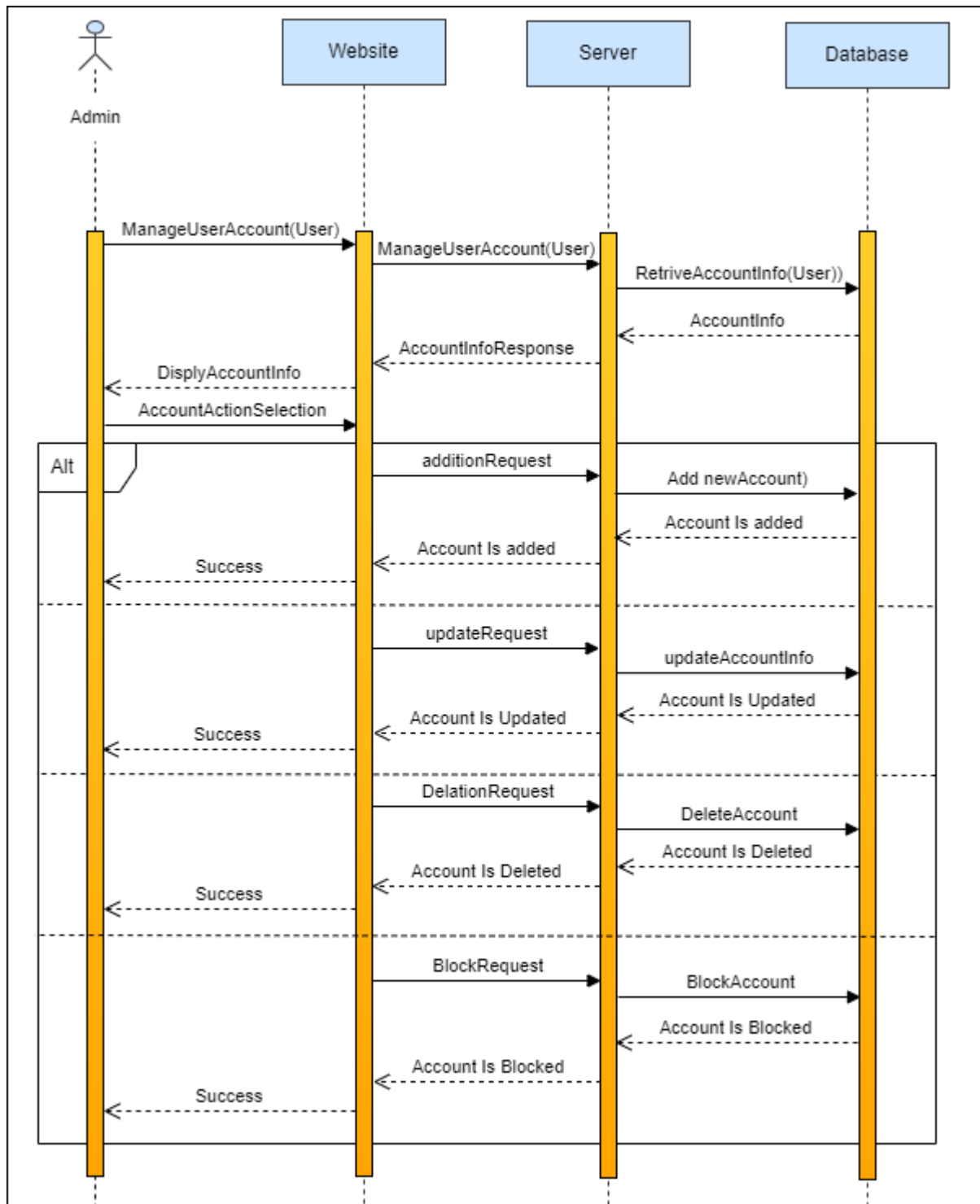


Figure 3.10. Manage users' account sequence diagram

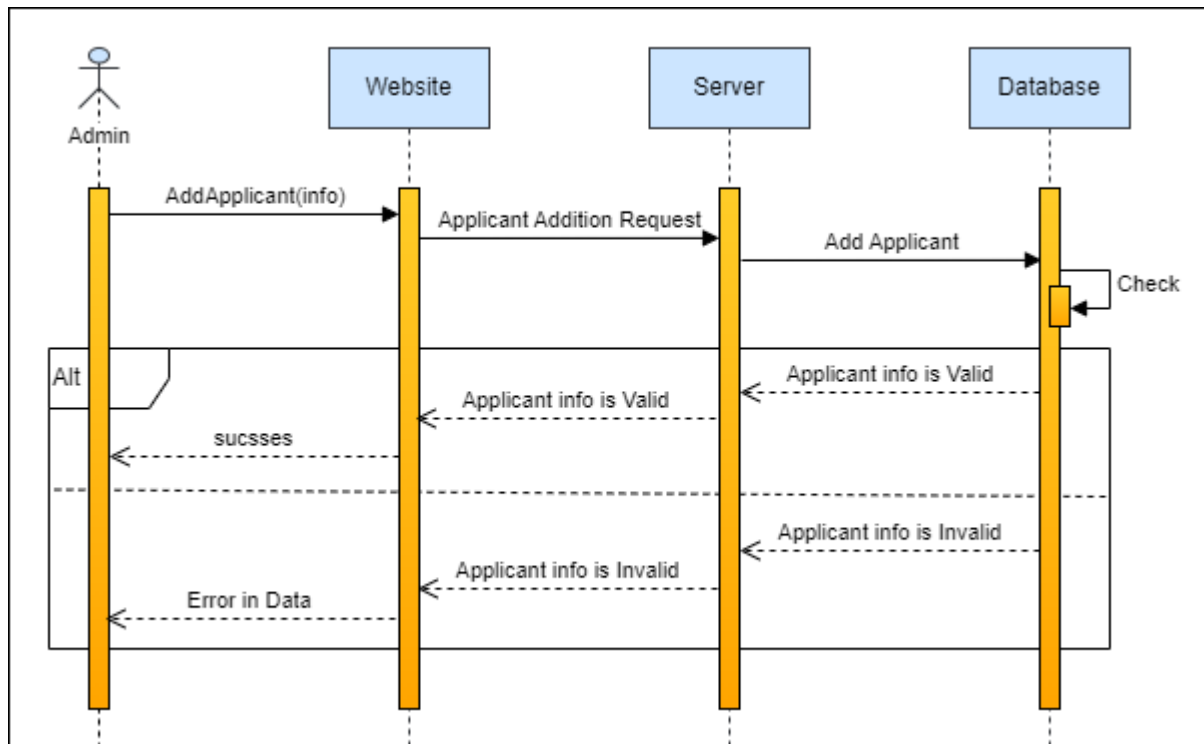


Figure 3.11. add Applicant sequence diagram

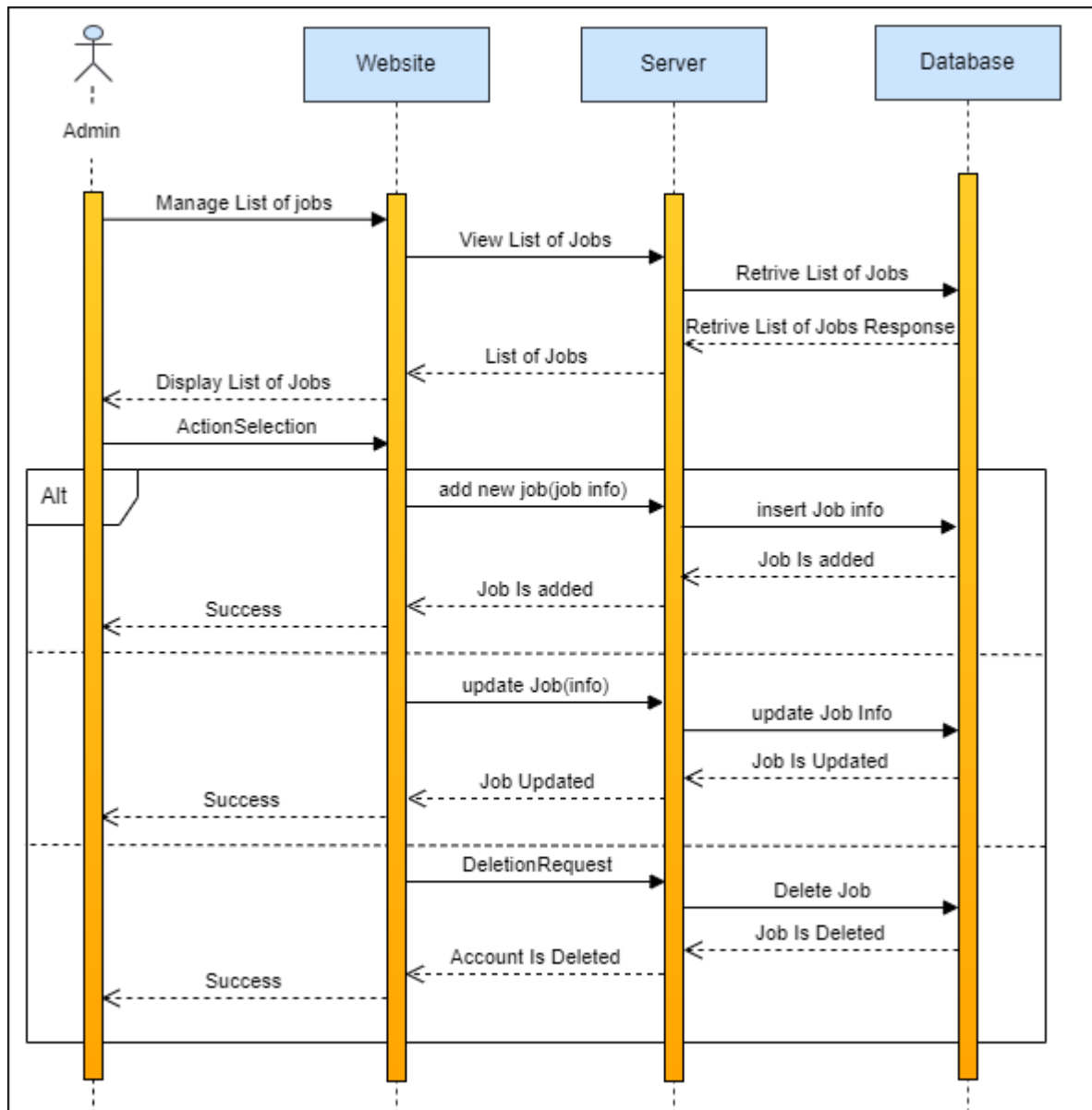


Figure 3.12. manage a list of jobs sequence diagram

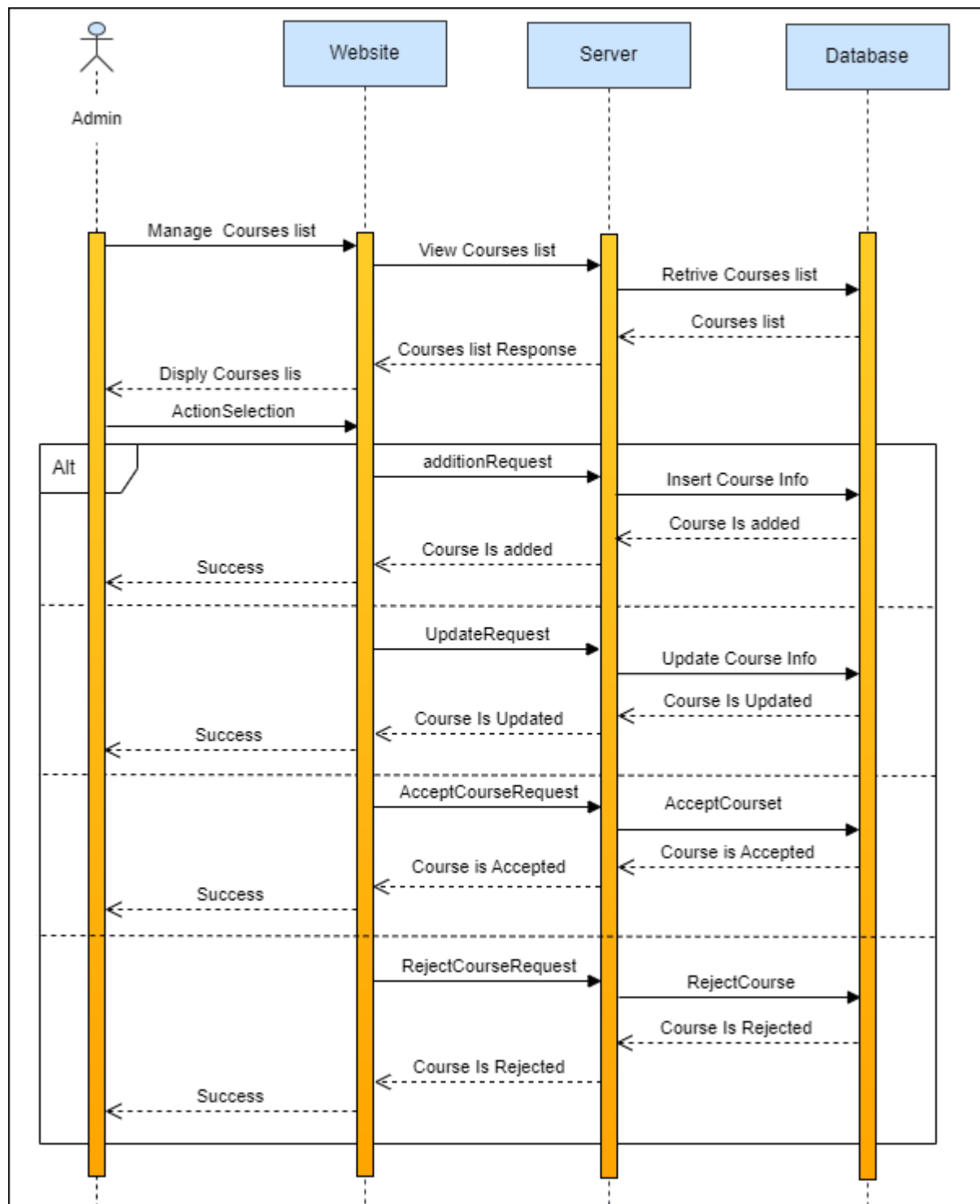


Figure 3.13. manage the list of Recommendation Courses sequence diagram

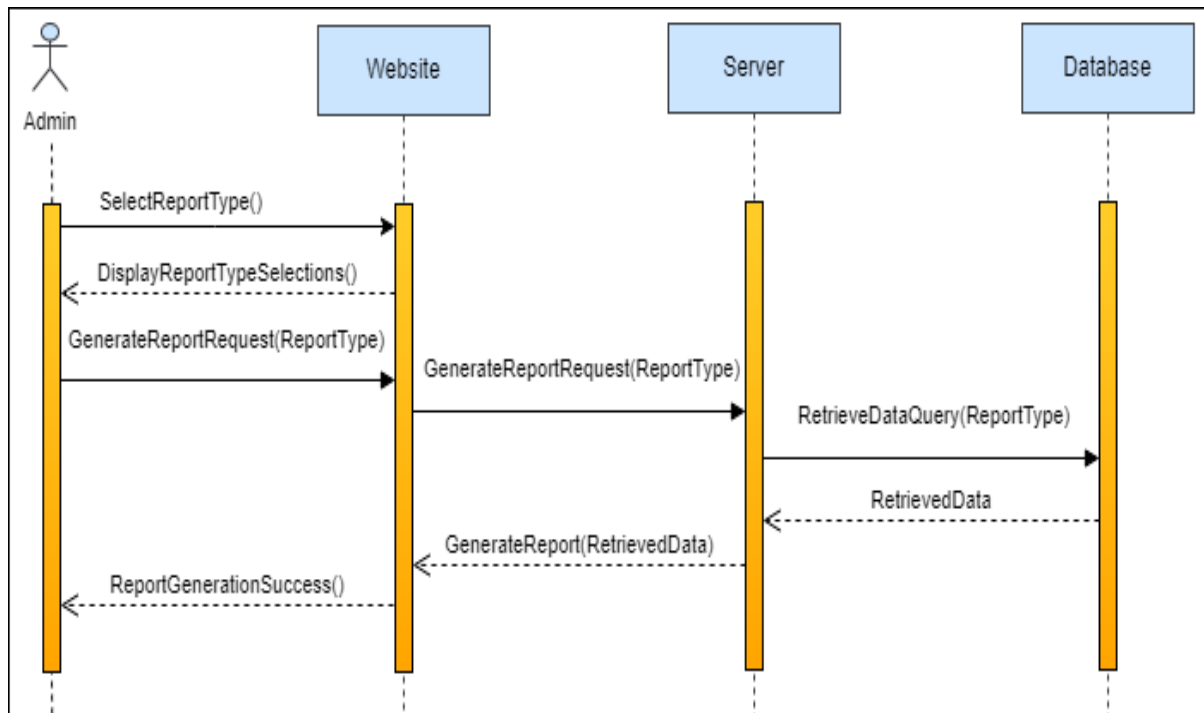


Figure 3.14. Generate Reports sequence diagram

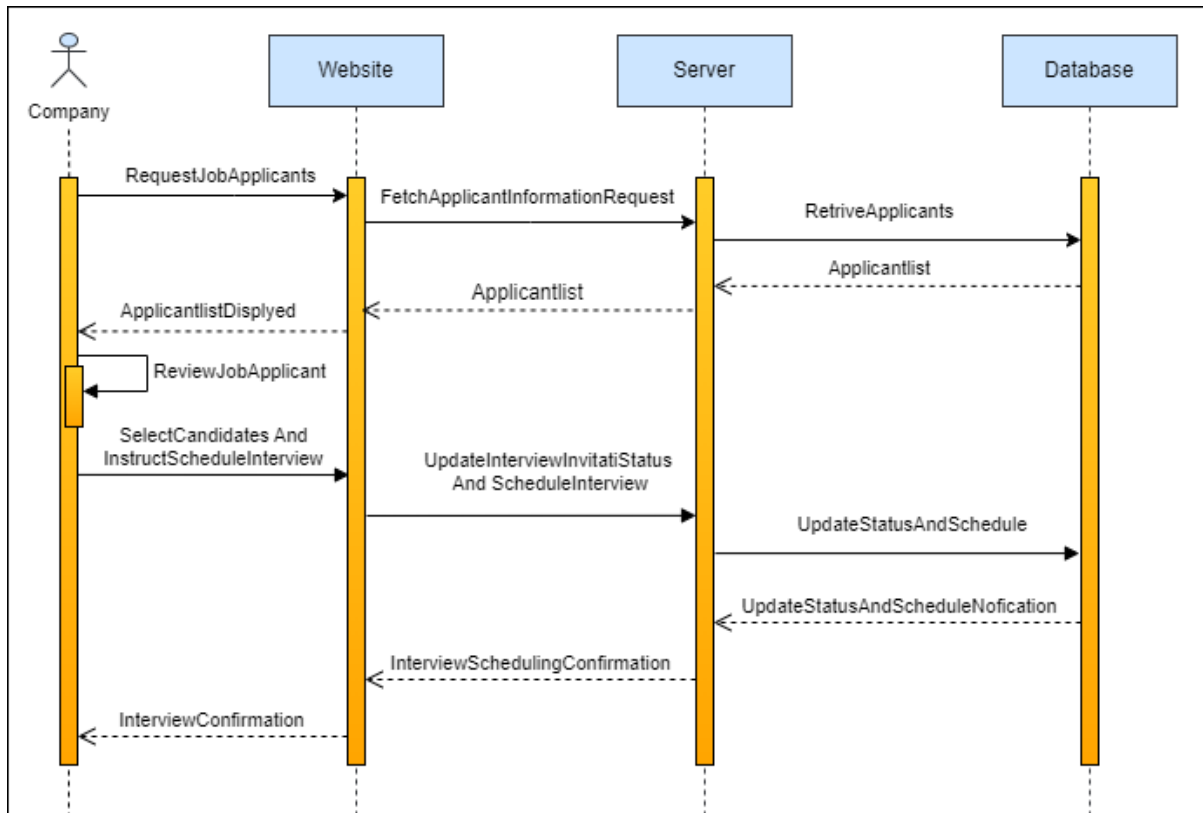


Figure 3.15. Company - View Job Applicants and Select Candidates and Request Interview

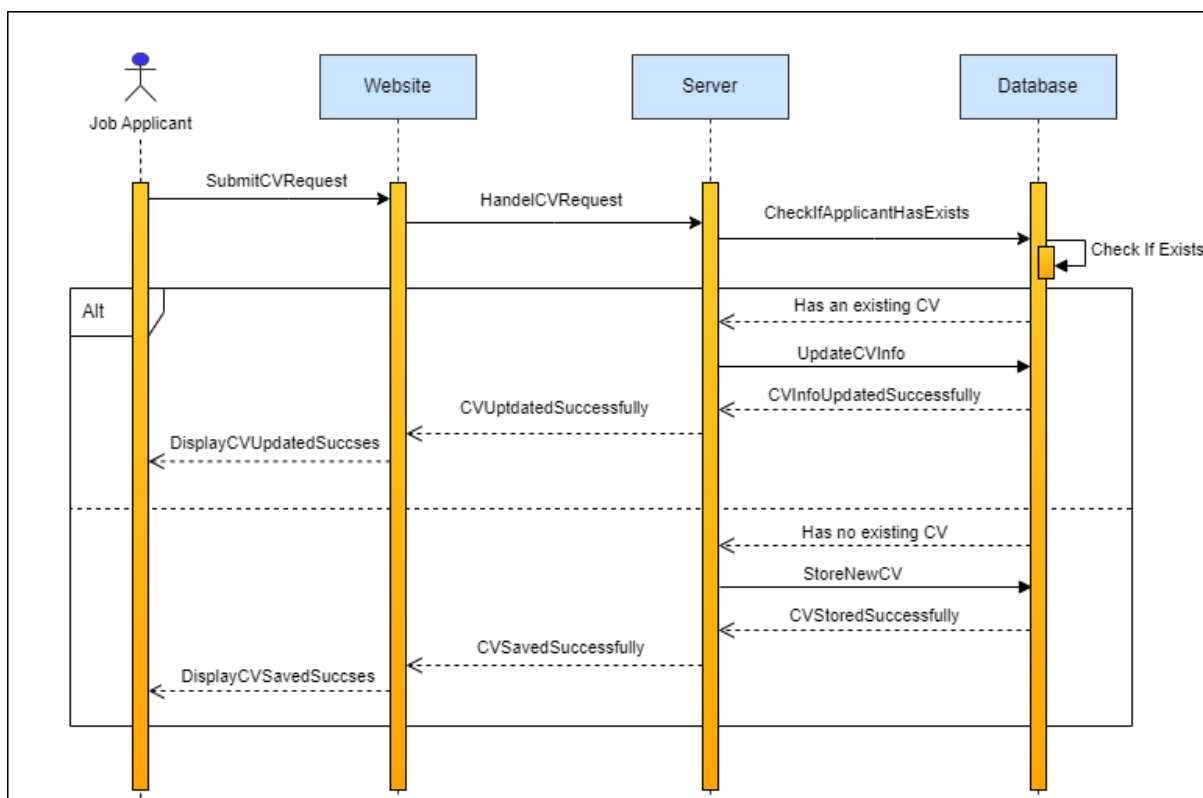


Figure 3.16. Job applicant- Add/Update CV sequence diagram

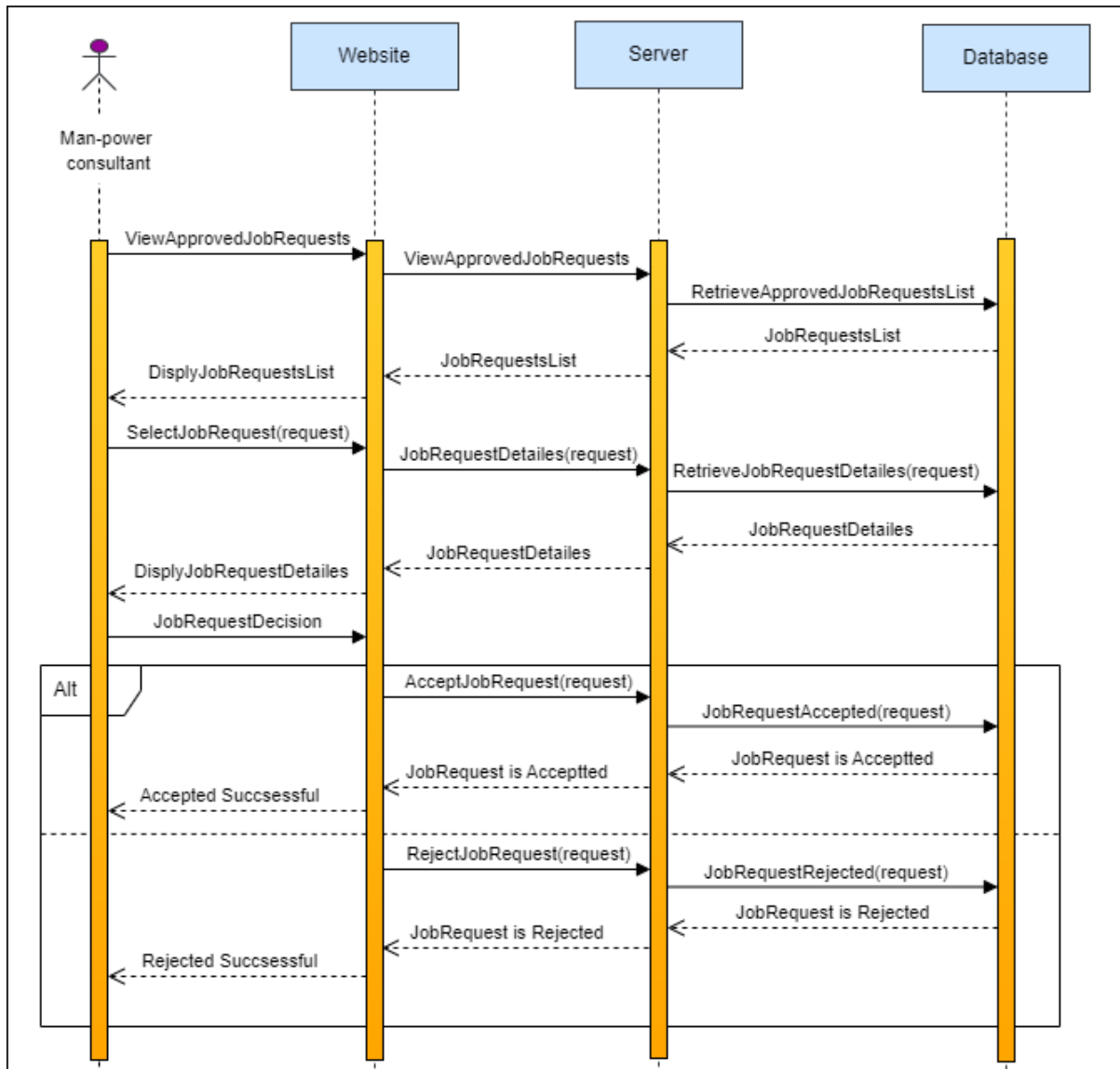


Figure 3.17. View Approved Job Request/accept/reject sequence diagram

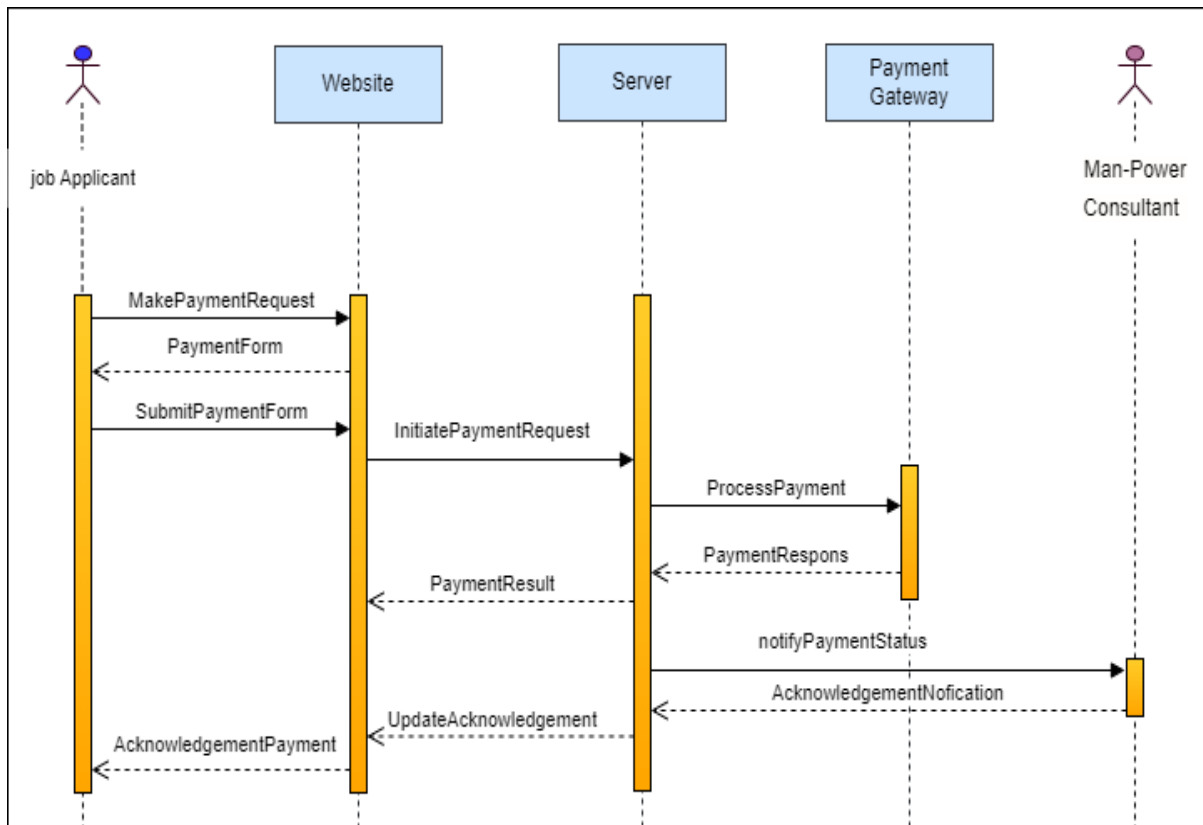


Figure 3.18. Make a Payment sequence diagram

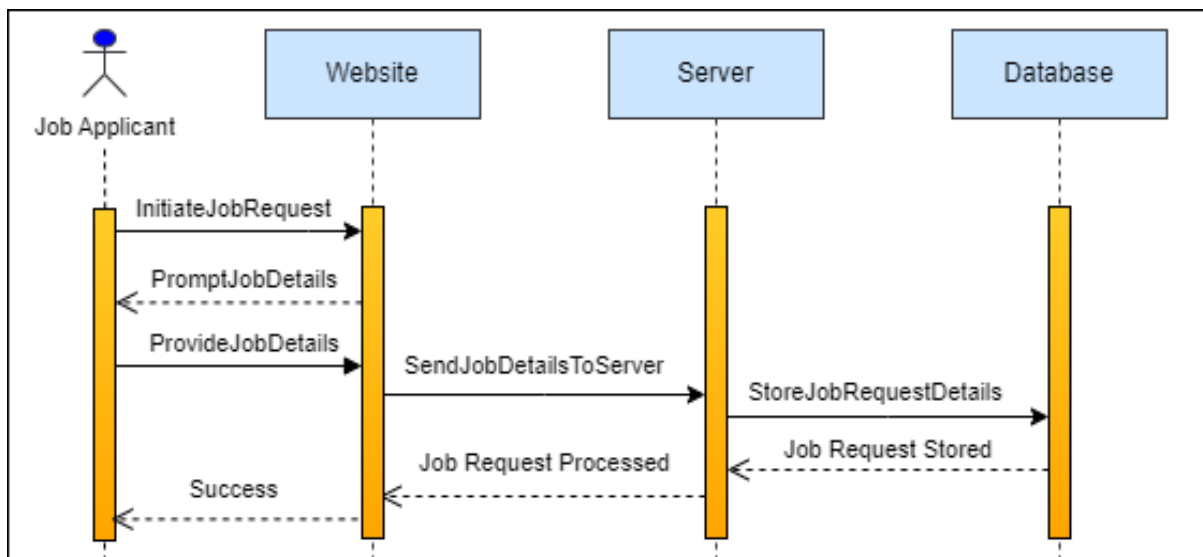


Figure 3.19. Job applicant- Send Job Request sequence diagram

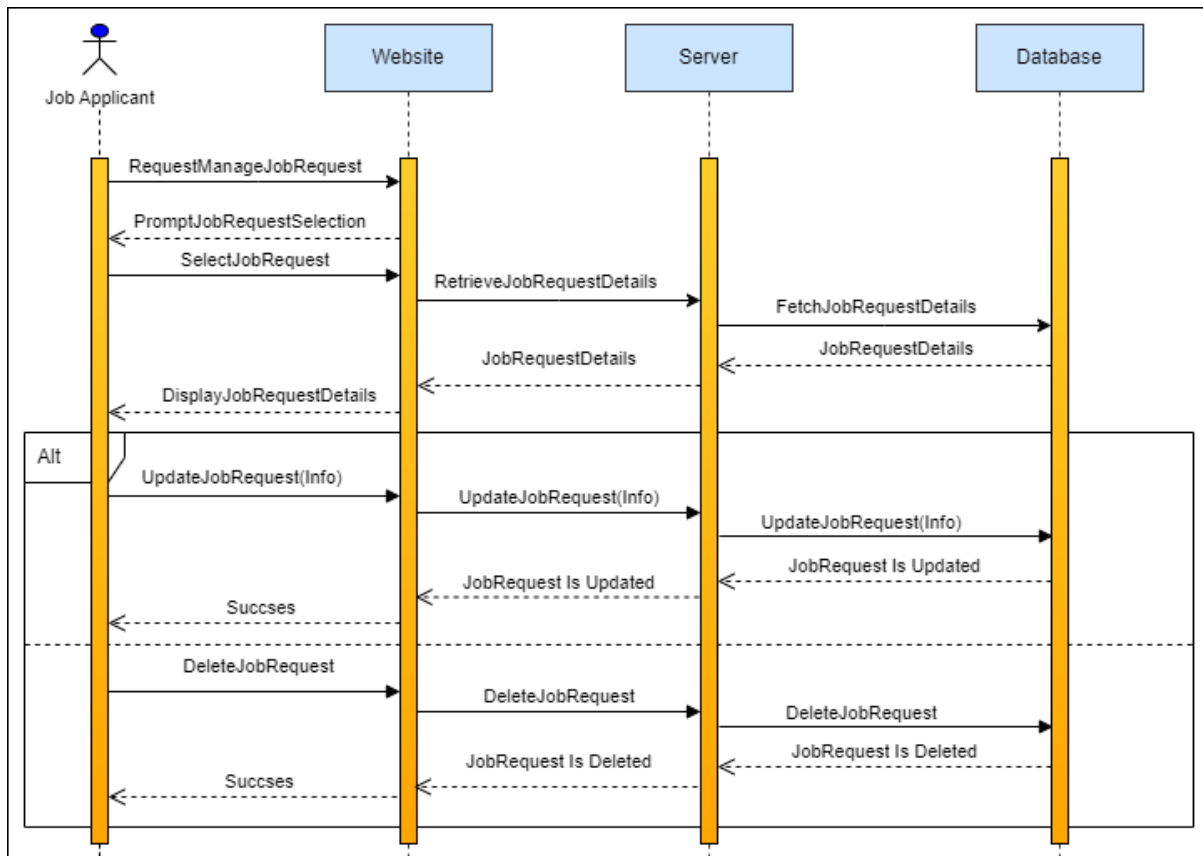


Figure 3.20. Job applicant- manage Job Request sequence diagram

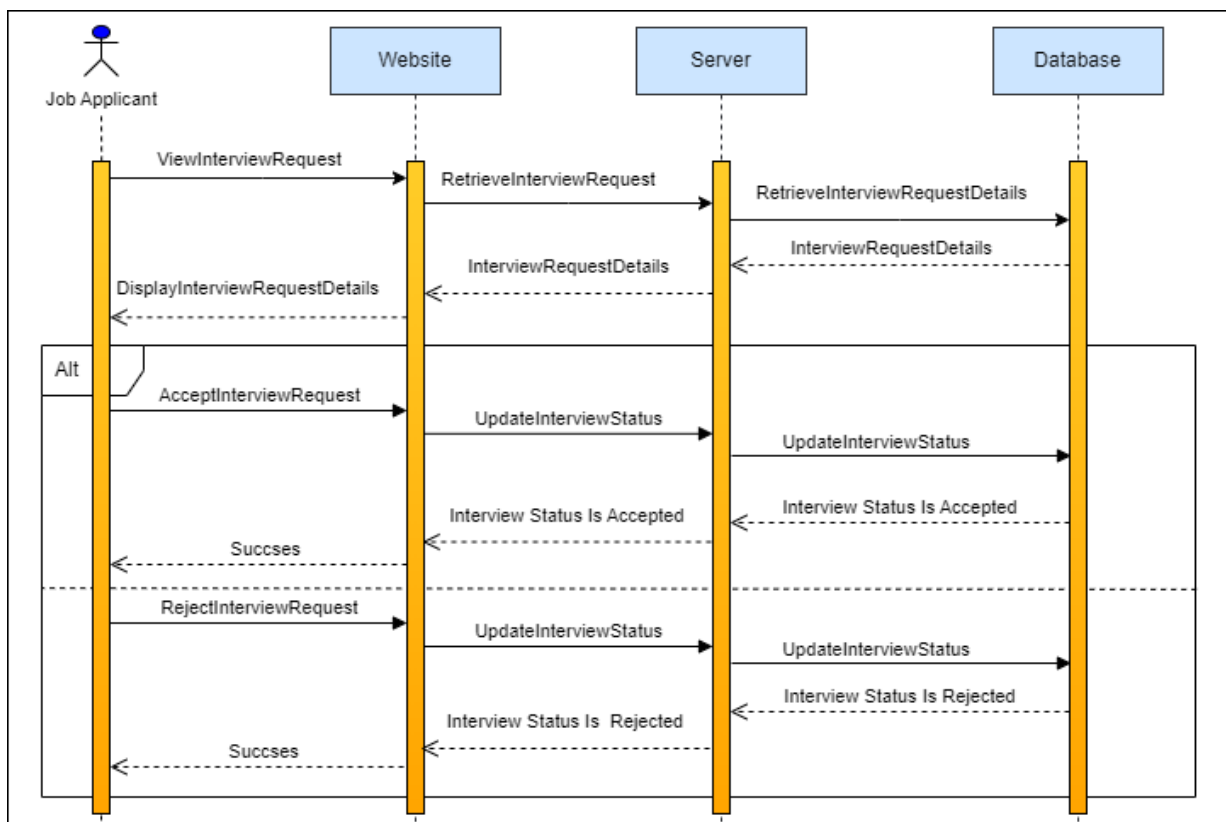


Figure 3.21. View Interview Request and Accept/Reject

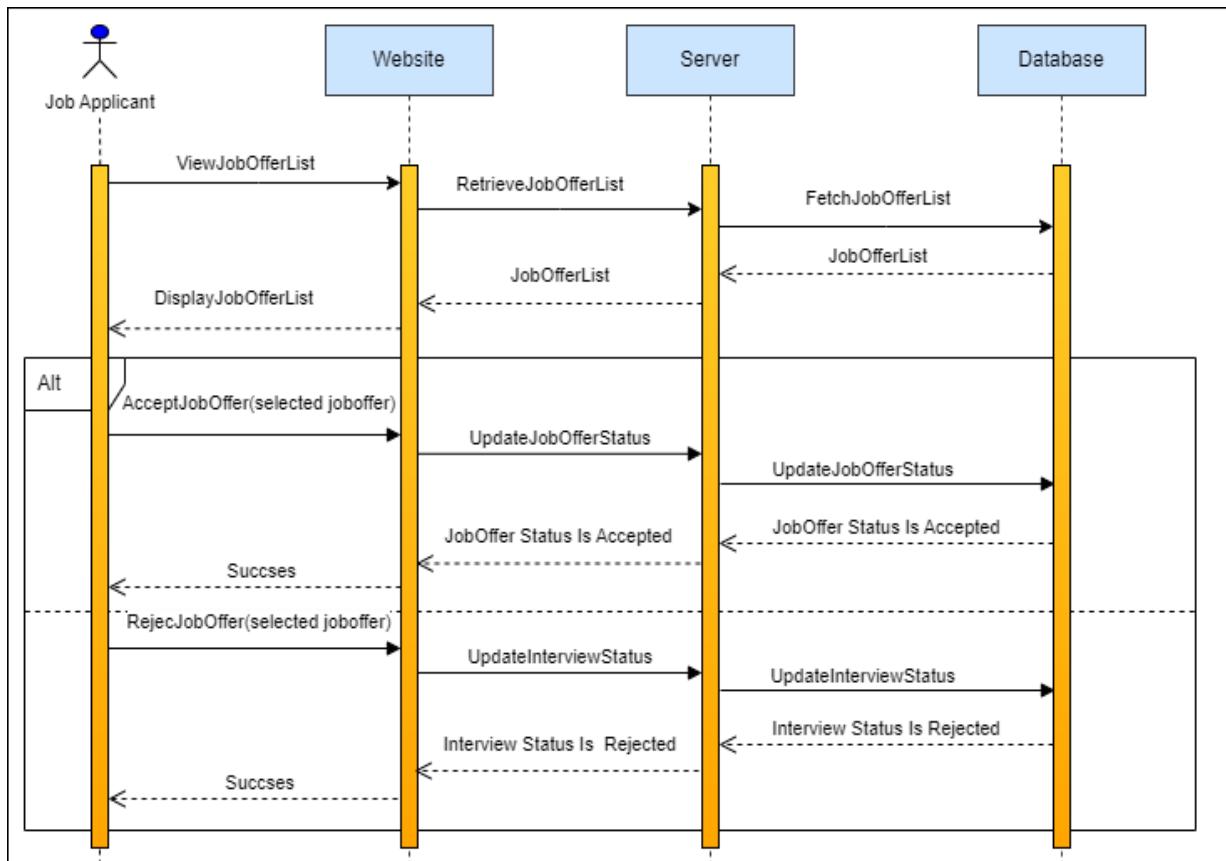


Figure 3.22. View the list of job offer and Accept/Reject

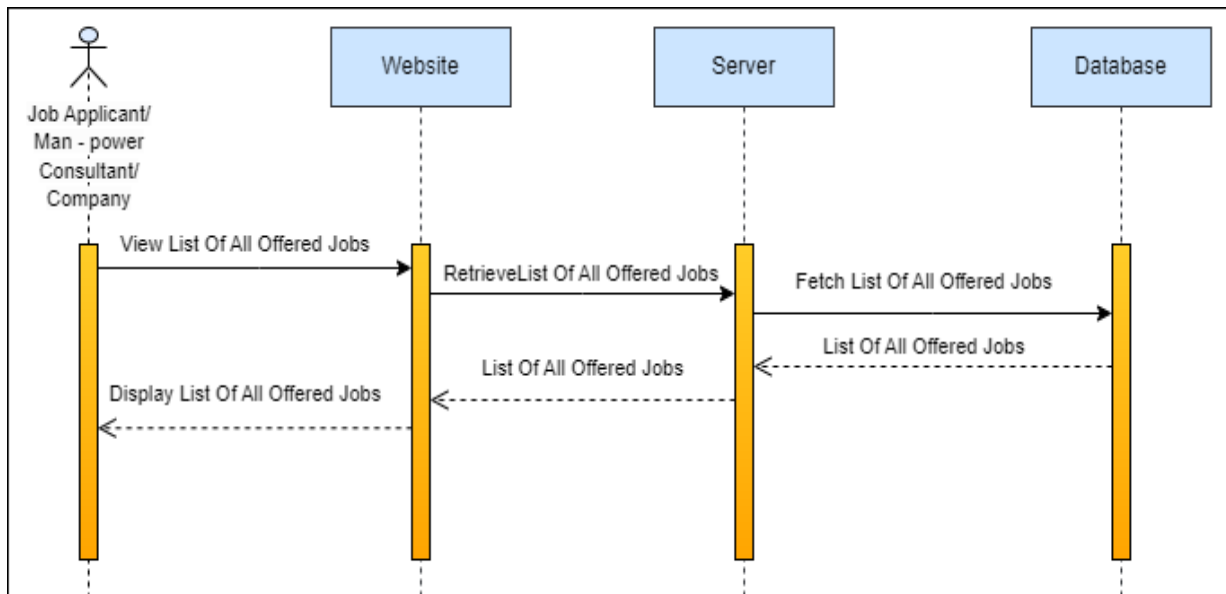


Figure 3.23. View the list of all offered jobs sequence diagram

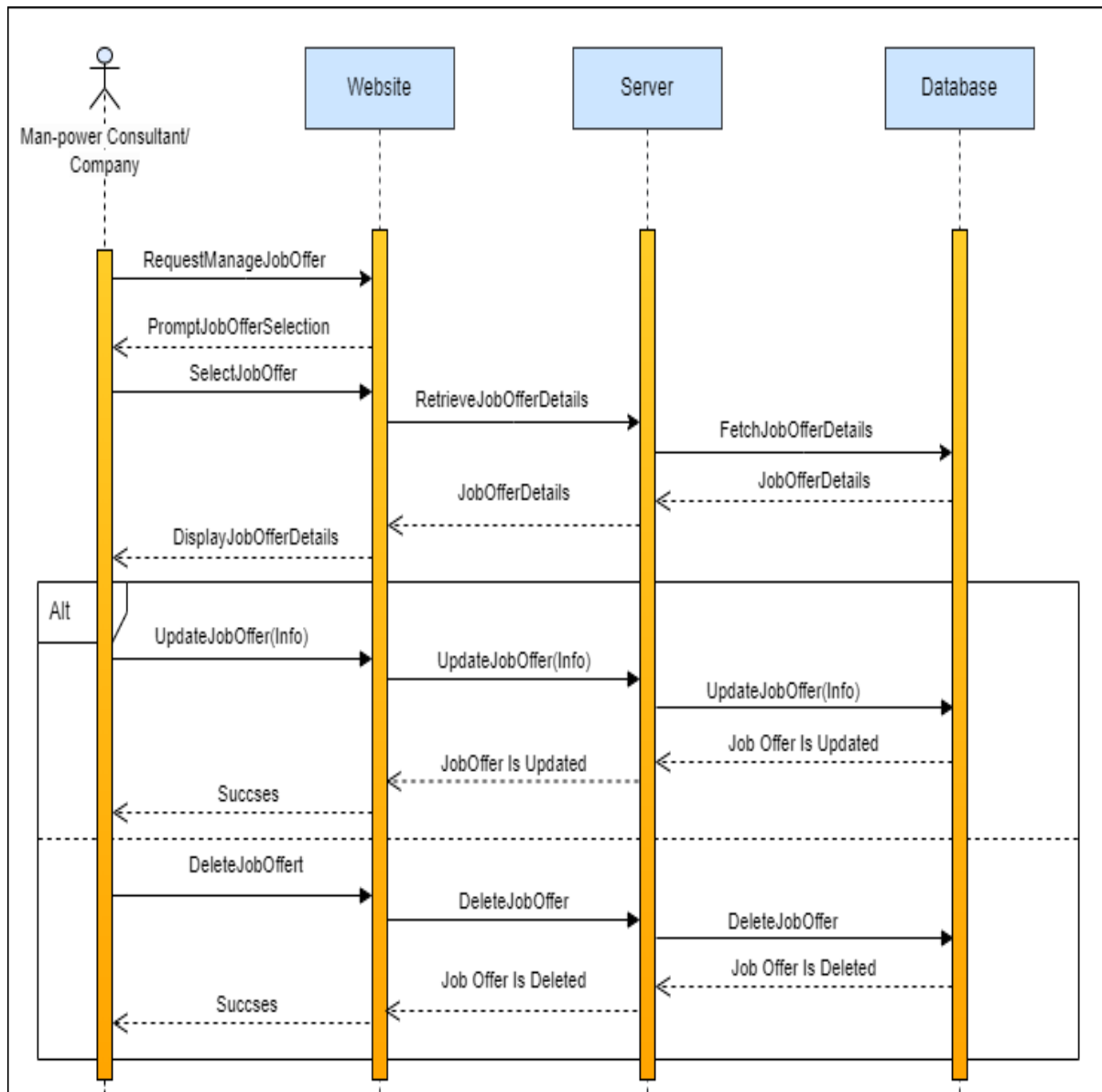


Figure 3.24. manage job offers sequence diagram

3.5.3 Activity Diagram

An activity diagram is employed to depict the progression of messages between different activities. Its main purpose is to capture the dynamic behavior of the system, showcasing its active state. Activity diagrams play a crucial role in the construction of executable systems through forward and reverse engineering approaches, while also providing insights into the dynamic nature of the system [14].

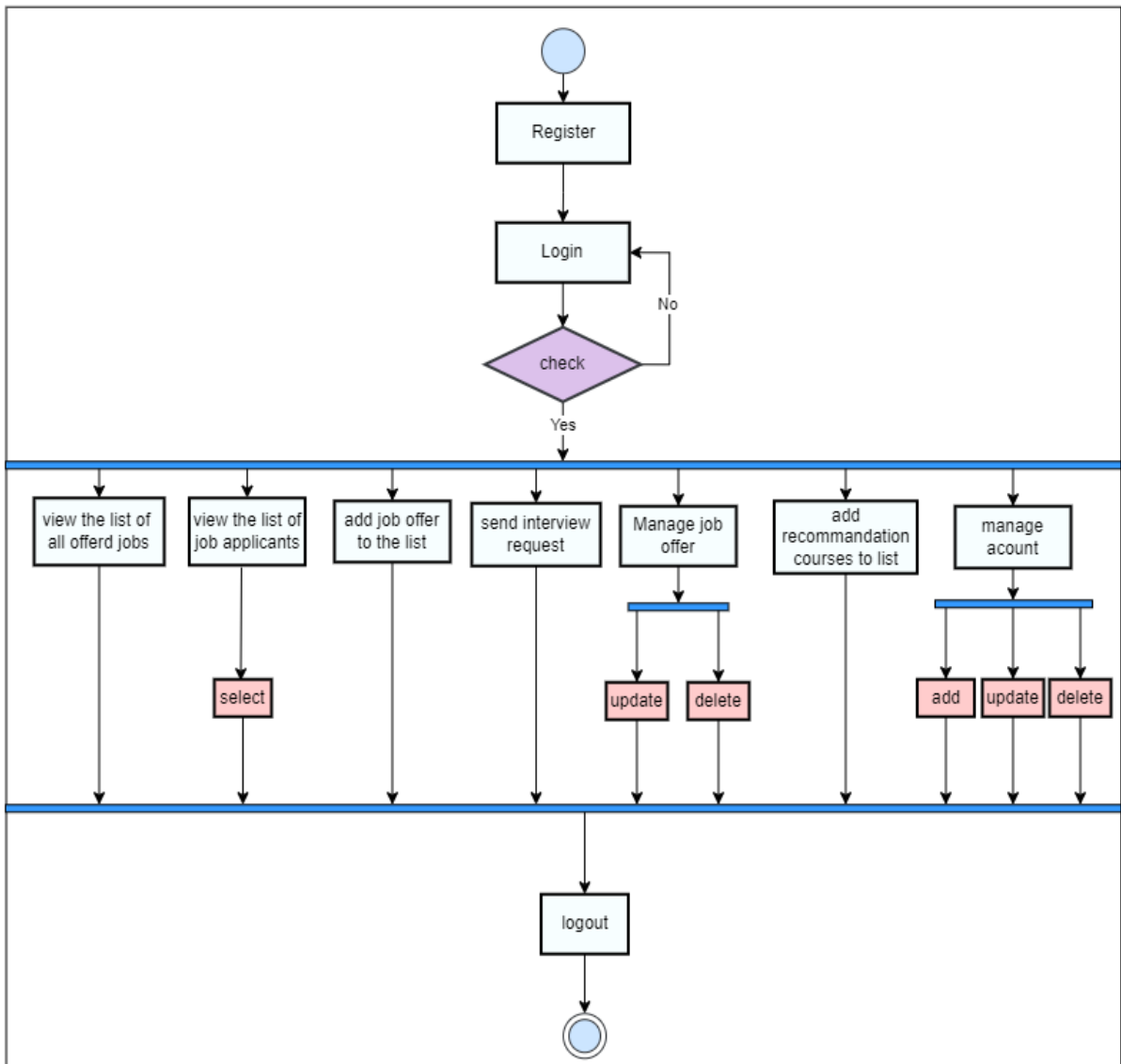


Figure 3.25. Company activity diagram

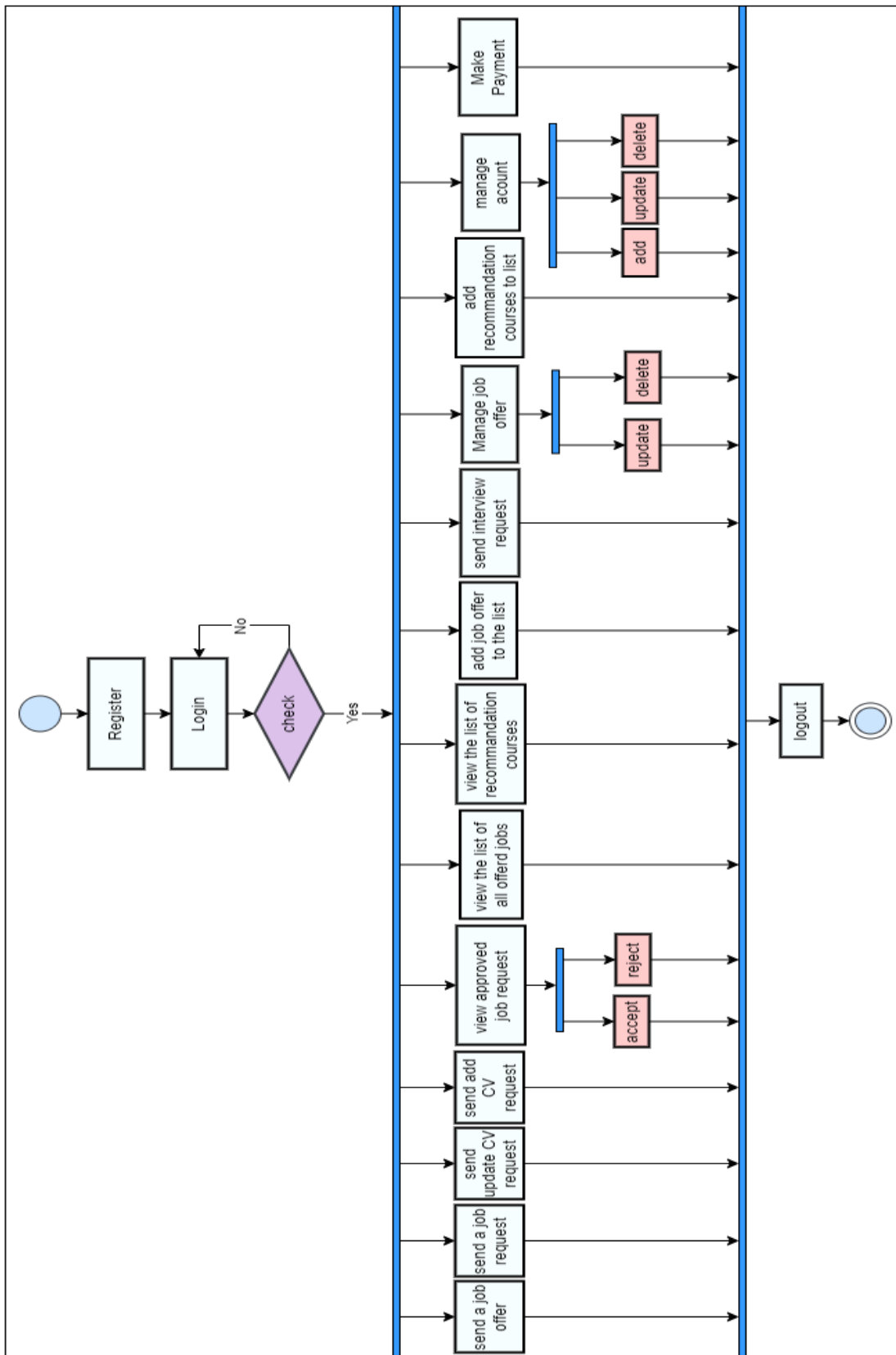


Figure 3.26. Man-power consultant activity diagram

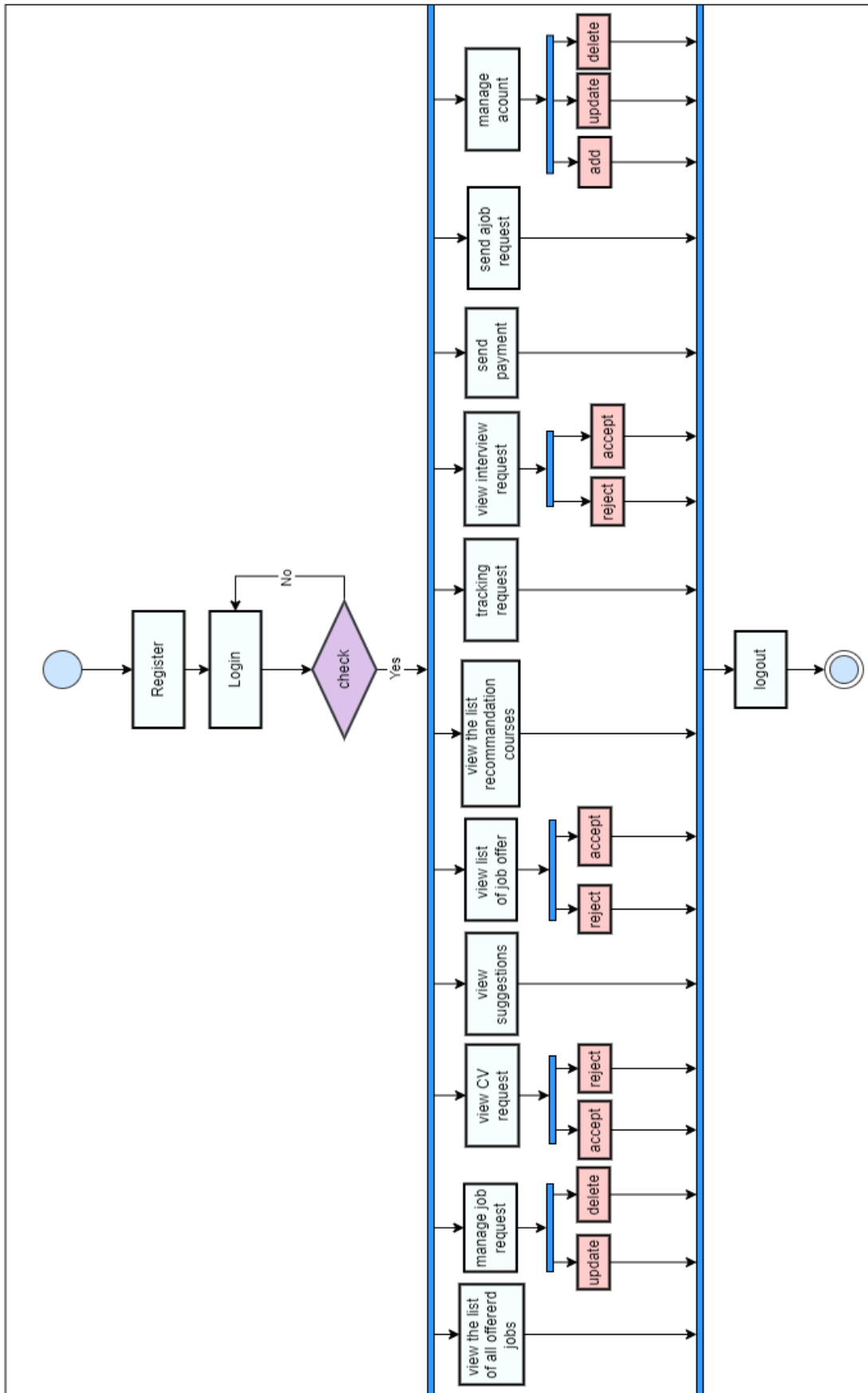


Figure 3.27. Job Applicant activity diagram

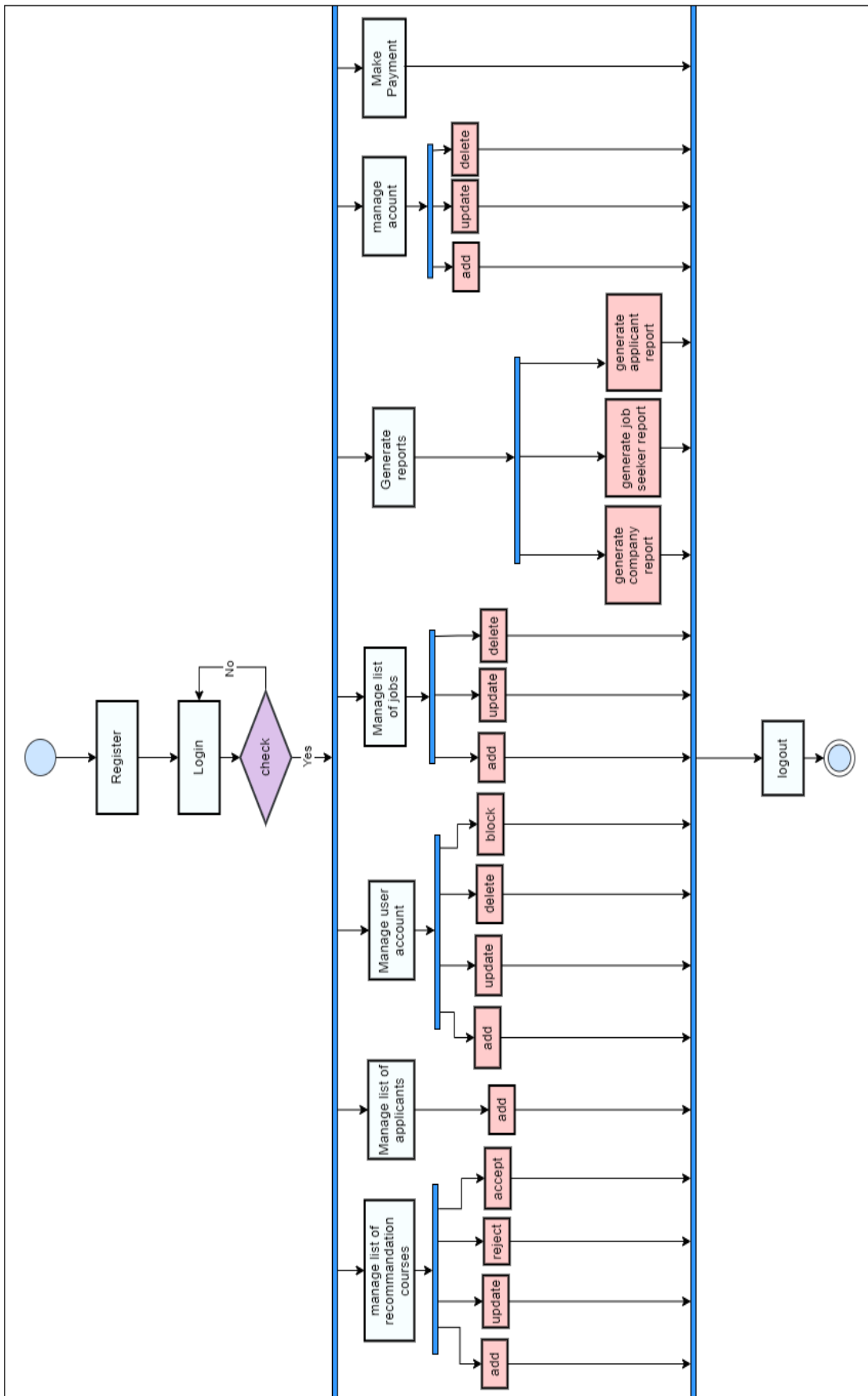


Figure 3.28. Admin activity diagram

3.5.4 Dataflow Diagram DFD

A data flow diagram (DFD) visually represents the inputs and outputs involved in the data processing of a system. It centers around the flow of info centers depicting the origin of data, its path, and how it is stored [15].

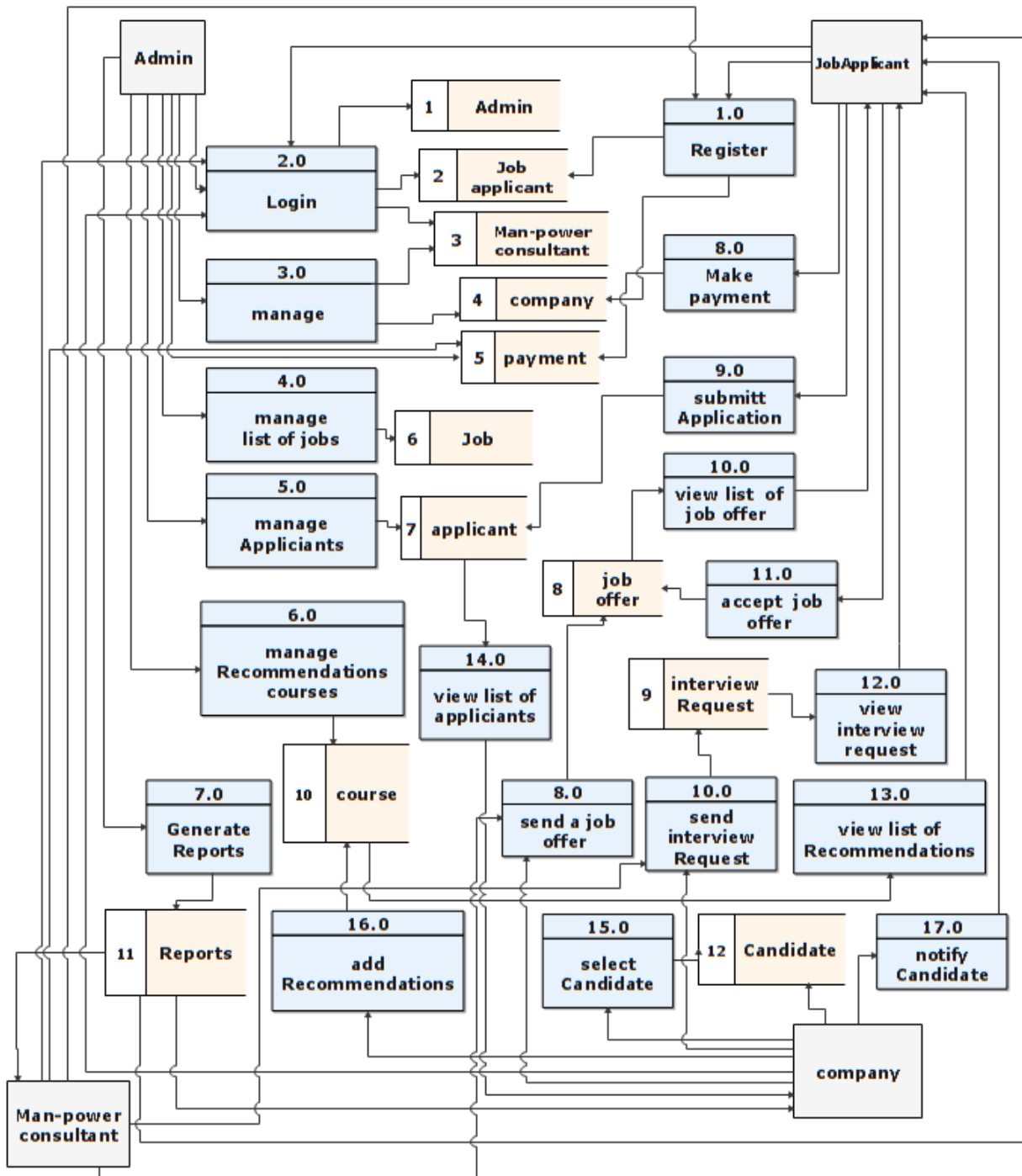


Figure 3.29. DFD diagram

3.5.5 Entity Relationship Diagram (ERD)

An entity relationship model (ERM), also known as an entity relationship diagram (ERD), is a visual depiction that illustrates the connections among entities, objects, places, concepts, or occurrences within an information technology system. It serves to define business processes and provide a foundation for a relational database [16].

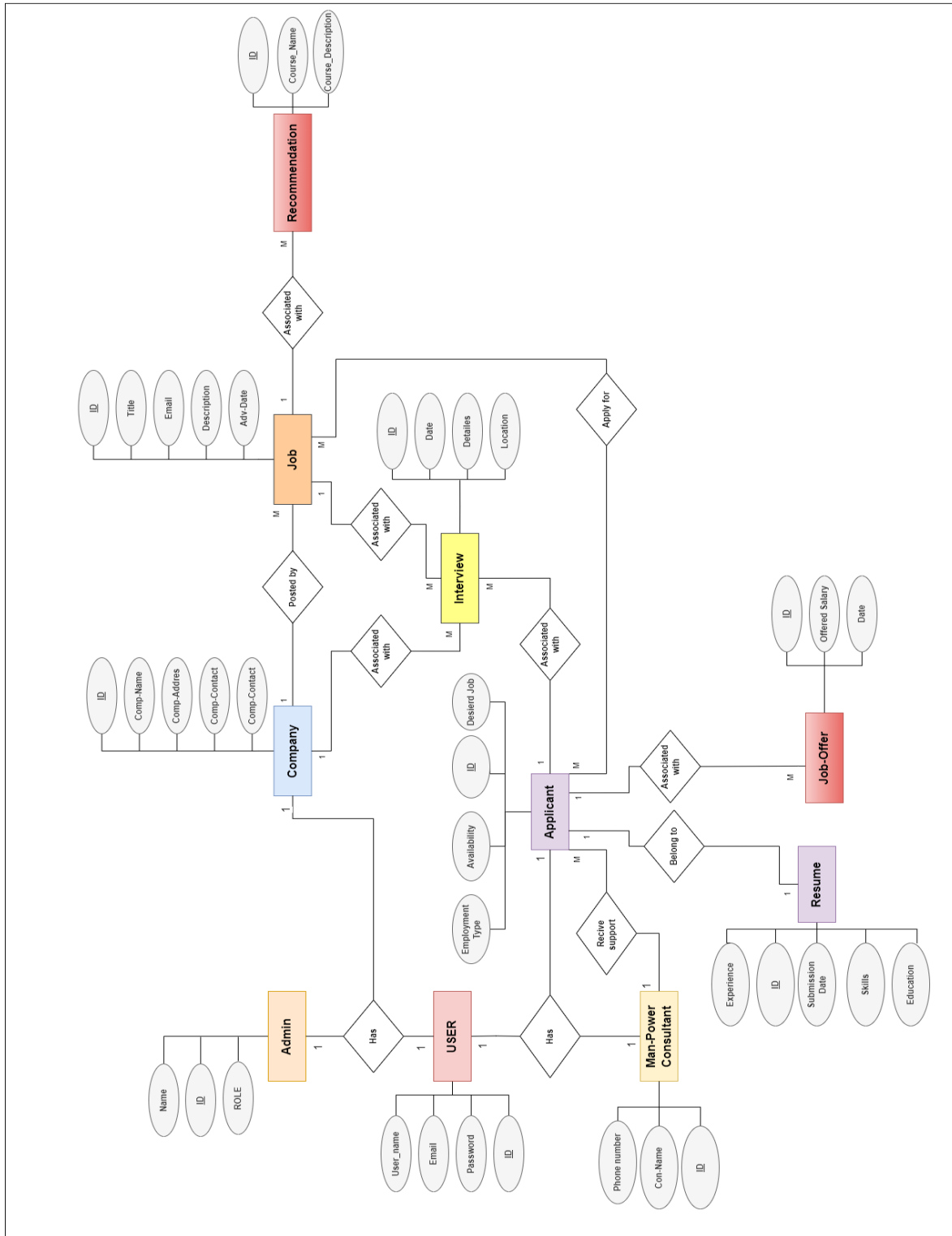


Figure 3.30. Entity Relation Diagram

3.5.6 Class Diagram

Class diagrams are highly valuable diagrams in UML as they effectively illustrate the organization of a particular system by representing its classes, attributes, methods, and connections among objects [17].

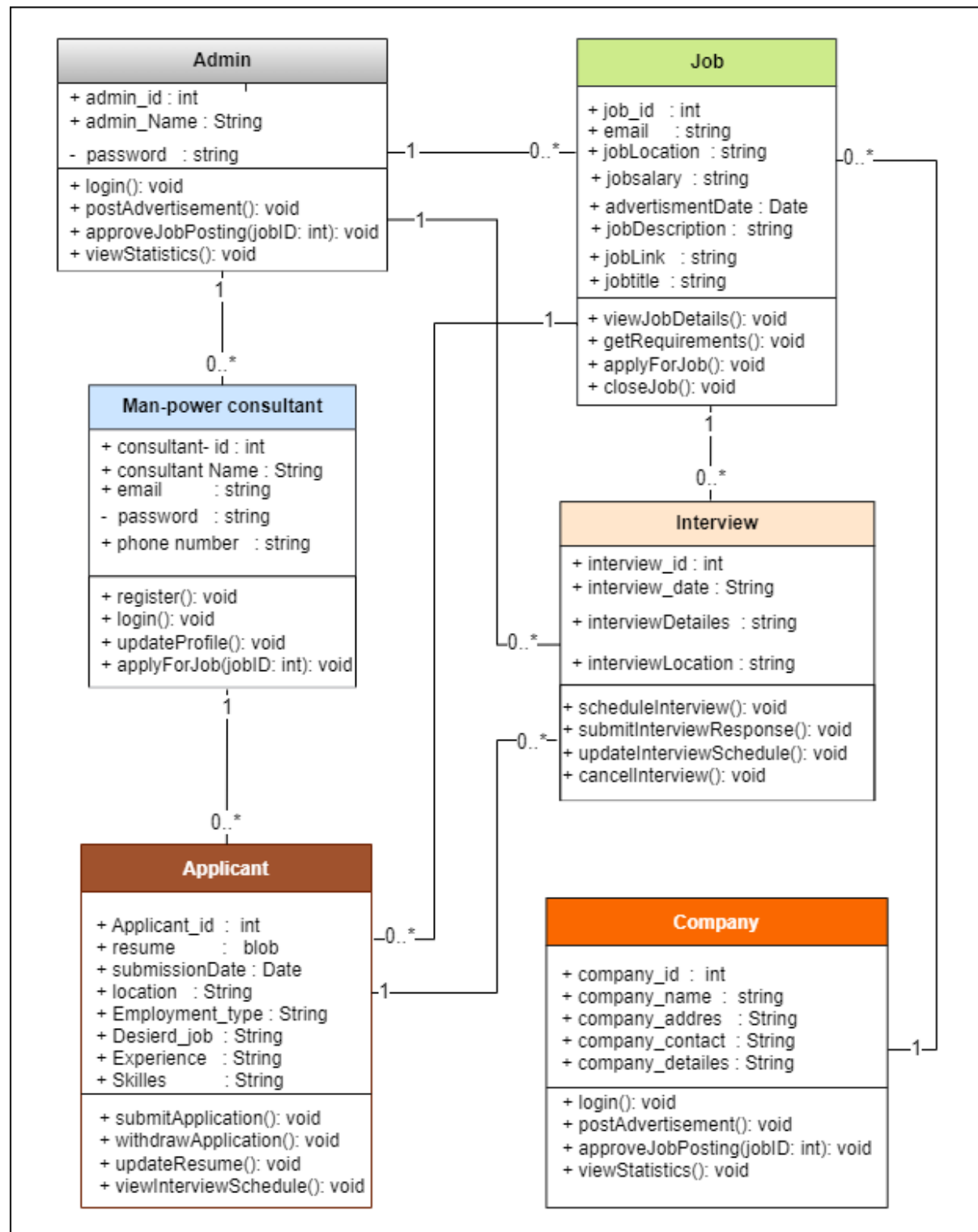


Figure 3.31. Class Diagram

3.5.7 Deployment Diagram

Deployment diagrams represent the physical structure of a system, depicting how the processing components are distributed. These diagrams showcase the relationships between the software and hardware elements of the system, providing an overview of the system's physical architecture [18].

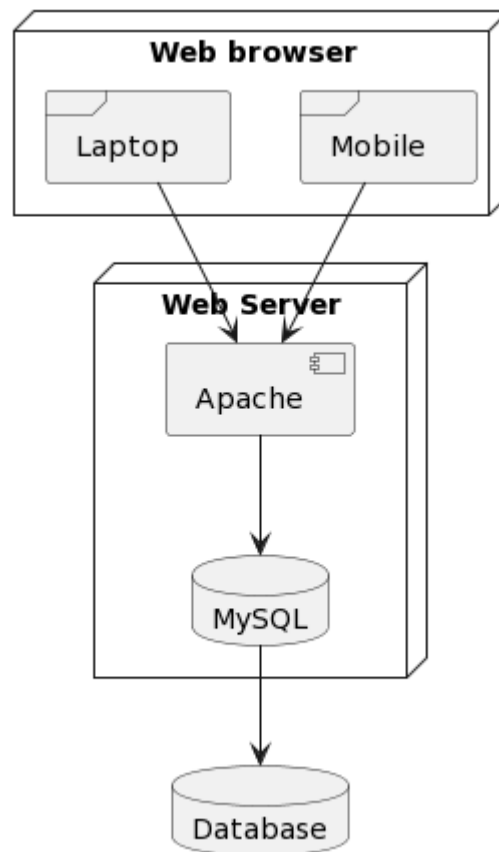
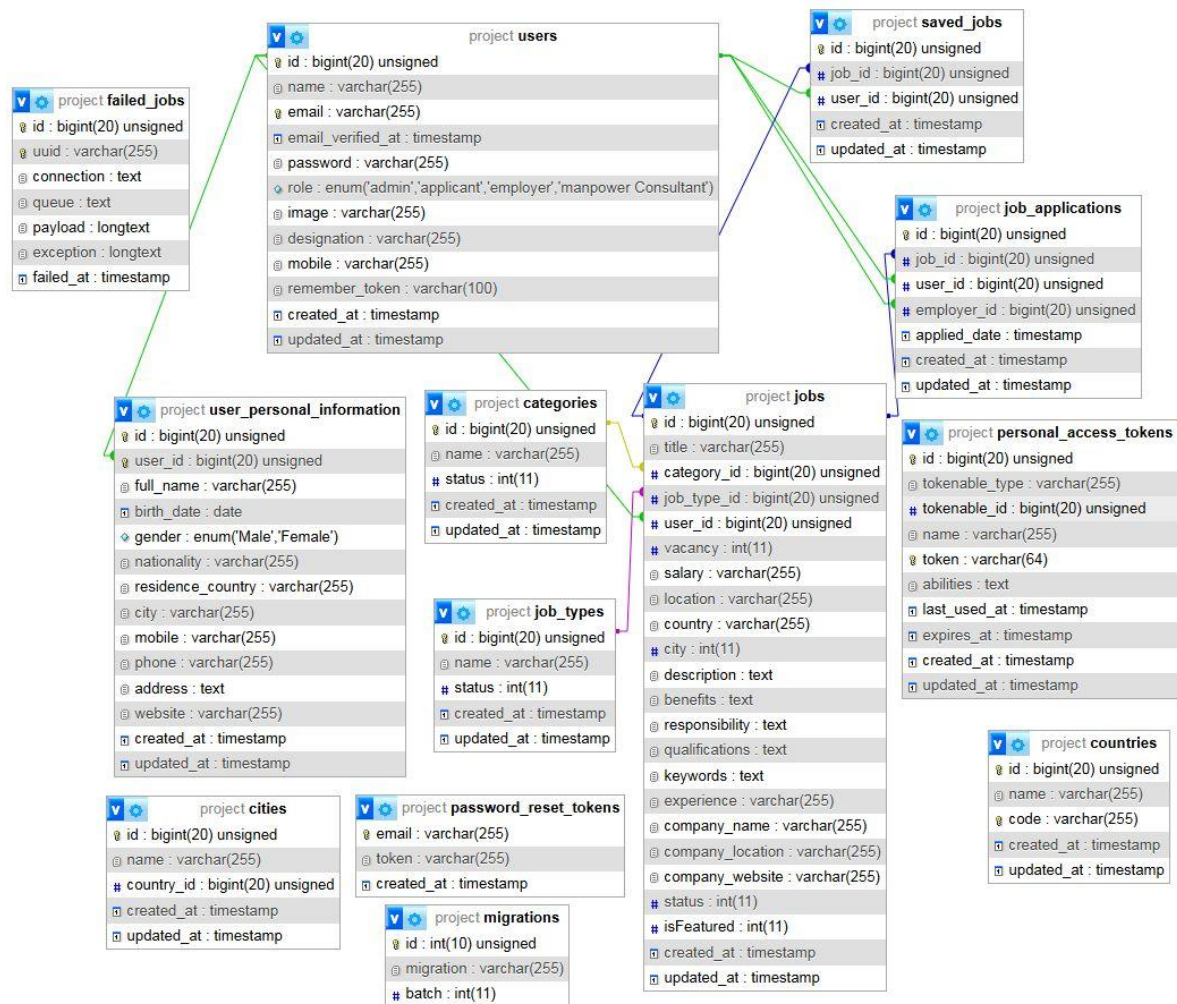


Figure 3.32. Deployment Diagram

3.6 Database Details

Database physical schema:



3.6.1 Create Database:

CREATE DATABASE your_database_name;

3.6.2 Create users Table:

```
CREATE TABLE `users` (
  `id` bigint(20) UNSIGNED NOT NULL,
  `name` varchar(255) NOT NULL,
  `email` varchar(255) NOT NULL,
```

`password` varchar(255) NOT NULL)

3.6.3 Create Admin Table:

```
CREATE TABLE `admin` (  
  `Admin_Id` int(11) NOT NULL,  
  `Name` varchar(50) NOT NULL,  
  `Role` varchar(50) NOT NULL,  
  `user_id` bigint(20) UNSIGNED NOT NULL  
)
```

3.6.4 Create Applicant Table:

```
CREATE TABLE `applicant` (  
  `Applicant_id` int(11) NOT NULL,  
  `Applicant_Name` varchar(100) NOT NULL,  
  `Employment_type` tinyint(4) NOT NULL,  
  `Consultant_Id` int(11) DEFAULT NULL,  
  `Availability` tinyint(4) NOT NULL,  
  `Desired_job` varchar(50) DEFAULT NULL,  
  `user-id` bigint(20) UNSIGNED NOT NULL  
)
```

3.6.5 Create Application Table:

```
CREATE TABLE `application` (  
  `Application_Id` int(11) NOT NULL,  
  `Application_date` date NOT NULL,  
  `Applicant_Id` int(11) NOT NULL,  
  `Job_Id` int(11) NOT NULL,  
  `Application_Status` varchar(50) NOT NULL  
)
```

3.6.6 Create Company Table:

```
CREATE TABLE `company` (  
  `Company_Id` int(11) NOT NULL,  
  `Company_Name` varchar(255) NOT NULL,  
  `Company_address` varchar(255) DEFAULT NULL,  
  `Company_Contact` varchar(50) DEFAULT NULL,
```

```
`Company_Details` text DEFAULT NULL,  
`user-id` bigint(20) UNSIGNED NOT NULL  
)
```

3.6.7 Create Interview Table:

```
CREATE TABLE `interview` (  
  `Interview_Id` int(11) NOT NULL,  
  `Applicant_Id` int(11) NOT NULL,  
  `Job_id` int(11) NOT NULL,  
  `Interview_Date` date NOT NULL,  
  `Interview_Details` text DEFAULT NULL,  
  `Interview_Location` varchar(255) DEFAULT NULL,  
  `Company_Id` int(11) NOT NULL  
)
```

3.6.8 Create Job Table:

```
CREATE TABLE `job` (  
  `Job_Id` int(11) NOT NULL,  
  `Job_Title` varchar(255) NOT NULL,  
  `JobDescription` text NOT NULL,  
  `Company_Id` int(11) NOT NULL,  
  `Adv_Date` date NOT NULL,  
  `Email` varchar(100) DEFAULT NULL,  
  `Job_Link` varchar(255) DEFAULT NULL  
)
```

3.6.9 Create Job_offer Table:

```
CREATE TABLE `job_offer` (  
  `Offer_Id` int(11) NOT NULL,  
  `Offered_Salary` decimal(10,2) NOT NULL,  
  `Date` date NOT NULL,  
  `Applicant_Id` int(11) NOT NULL  
)
```

3.6.10 Create Manpower_Consultant Table:

```
CREATE TABLE `manpower_consultant` (  

```

```

`Consultant_Id` int(11) NOT NULL,
,
`user-id` bigint(20) UNSIGNED NOT NULL
)

```

3.6.11 Create Recommendation Table:

```

CREATE TABLE `recommendation` (
  `Course_Id` int(11) NOT NULL,
  `Course_Name` varchar(255) NOT NULL,
  `Course_Description` text DEFAULT NULL,
  `Job_Id` int(11) NOT NULL
)

```

3.6.12 Create Resume Table:

```

CREATE TABLE `resume` (
  `Resume_Id` int(11) NOT NULL,
  `Submission_Date` date NOT NULL,
  `Skills` text DEFAULT NULL,
  `Experience` text DEFAULT NULL,
  `Education` text DEFAULT NULL,
  `Applicant_Id` int(11) NOT NULL
)

```

3.6.13 Constraints for users Table:

```

MODIFY `id` bigint(20) UNSIGNED NOT NULL AUTO_INCREMENT;

```

3.6.14 Constraints for Admin Table:

```

ALTER TABLE `admin`
  ADD CONSTRAINT `admin_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES `users` (`id`);
  MODIFY `Admin_Id` int(11) NOT NULL AUTO_INCREMENT;

```

3.6.15 Constraints for Applicant Table:

```

ALTER TABLE `applicant`
  ADD CONSTRAINT `USER-FK` FOREIGN KEY (`user-id`) REFERENCES `users` (`id`),
  ADD CONSTRAINT `applicant_ibfk_1` FOREIGN KEY (`Consultant_Id`) REFERENCES
`manpower_consultant` (`Consultant_Id`);
  MODIFY `Applicant_id` int(11) NOT NULL AUTO_INCREMENT;

```

3.6.16 Constraints for Application Table:

```
ADD CONSTRAINT `application_ibfk_1` FOREIGN KEY (`Applicant_Id`) REFERENCES `applicant` (`Applicant_id`),
```

```
ADD CONSTRAINT `application_ibfk_2` FOREIGN KEY (`Job_Id`) REFERENCES `job` (`Job_Id`);
```

```
MODIFY `Application_Id` int(11) NOT NULL AUTO_INCREMENT;
```

3.6.17 Constraints for Company Table:

```
ALTER TABLE `company`
```

```
ADD CONSTRAINT `user-id` FOREIGN KEY (`user-id`) REFERENCES `users` (`id`);
```

```
MODIFY `Company_Id` int(11) NOT NULL AUTO_INCREMENT;
```

3.6.18 Constraints for Interview Table:

```
ALTER TABLE `interview`
```

```
ADD CONSTRAINT `interview_ibfk_1` FOREIGN KEY (`Applicant_Id`) REFERENCES `applicant` (`Applicant_id`),
```

```
ADD CONSTRAINT `interview_ibfk_2` FOREIGN KEY (`Job_id`) REFERENCES `job` (`Job_Id`),
```

```
ADD CONSTRAINT `interview_ibfk_3` FOREIGN KEY (`Company_Id`) REFERENCES `company` (`Company_Id`);
```

```
MODIFY `Interview_Id` int(11) NOT NULL AUTO_INCREMENT;
```

3.6.19 Constraints for Job Table:

```
ALTER TABLE `job`
```

```
ADD CONSTRAINT `job_ibfk_1` FOREIGN KEY (`Company_Id`) REFERENCES `company` (`Company_Id`);
```

```
MODIFY `Job_Id` int(11) NOT NULL AUTO_INCREMENT;
```

3.6.20 Constraints for Job_offer Table:

```
ALTER TABLE `job_offer`
```

```
ADD CONSTRAINT `job_offer_ibfk_1` FOREIGN KEY (`Applicant_Id`) REFERENCES `applicant` (`Applicant_id`);
```

```
MODIFY `Offer_Id` int(11) NOT NULL AUTO_INCREMENT;
```

3.6.21 Constraints for Manpower_Consultant Table:

```
ALTER TABLE `manpower_consultant`
```

```
ADD CONSTRAINT `user-id1` FOREIGN KEY (`user-id`) REFERENCES `users` (`id`);
```

```
MODIFY `Consultant_Id` int(11) NOT NULL AUTO_INCREMENT;
```


3.6.22 Constraints for Recommendation Table:

```
ALTER TABLE `recommendation`
```

```
ADD CONSTRAINT `recommendation_ibfk_1` FOREIGN KEY (`Job_Id`) REFERENCES `job`  
(`Job_Id`);
```

```
MODIFY `Course_Id` int(11) NOT NULL AUTO_INCREMENT;
```

3.6.23 Constraints for Resume Table:

```
ALTER TABLE `resume`
```

```
ADD CONSTRAINT `resume_ibfk_1` FOREIGN KEY (`Applicant_Id`) REFERENCES `applicant`  
(`Applicant_id`);
```

```
MODIFY `Resume_Id` int(11) NOT NULL AUTO_INCREMENT;
```

3.7 Conclusion

In this chapter, we laid the groundwork for the emergency medical services website. By following the Systems Development Life Cycle (SDLC), conducting surveys, and creating UML diagrams, we've set the stage for successful implementation. The database design ensures data integrity. Next, we'll bring these designs to life.

CHAPTER FOUR: SYSTEM IMPLEMENTATION

1.1. Introduction

Chapter Four focuses on the implementation of the system. This chapter is a crucial phase in the project, where the system requirements are translated into a tangible and functional application. The implementation process involves converting design specifications into code, developing the user interface, and integrating the necessary functionalities.

The primary goal of system implementation is to transform the conceptual ideas and designs into a fully functional system that meets the users' needs. It requires careful planning, attention to detail, and technical expertise to ensure a successful implementation.

During the implementation phase, developers will utilize the Laravel framework, a powerful PHP framework known for its efficiency and robustness. Laravel provides a solid foundation for building web applications, offering features such as a modular architecture, database integration, and a wealth of pre-built components.

To begin the implementation process, it is essential to understand the requirements for a smooth development journey. This includes both hardware and software requirements. The hardware requirements ensure that the development environment has the necessary resources to support the implementation process effectively. On the other hand, the software requirements encompass the specific tools, frameworks, and libraries needed for successful implementation.

User interface (UI) and user experience (UX) are crucial aspects of the system implementation. The UI focuses on designing and creating visually appealing interfaces that are intuitive and user-friendly. The UX, on the other hand, concentrates on providing a seamless and satisfying experience for users as they navigate and interact with the system.

Front-end development is a critical component of the implementation process, as it involves implementing the user interface and handling client-side functionality. This includes designing and developing the visual elements, integrating responsive design principles, and enhancing user interactions using JavaScript and AJAX.

1.2. Requirments

4.2.1. Hardware

Table 4.1 : Hardware

Device	Specification
CPU	Laptop with Core i7
RAM	8 GB
Storage Space	128 GB SSD
Operating System	Windows 10 or macOS Mojave

4.2.2. Software Requirements:

Table 4.2 :Software

Front-end	HTML, CSS, JavaScript,
Back-end	PHP, Laravel framework
Database	MySQL (8.0)

4.3 Implementation Steps

4.3.1 User Interface (UI) and UX (User Experience)

Our focus during the implementation phase was to create a user interface (UI) that not only exuded a sleek and modern aesthetic but also provided a seamless and intuitive user experience (UX). To achieve this, we followed a comprehensive and iterative approach, incorporating the following steps:

1. **Unveiling the Users:** We began by conducting extensive user research to gain a deep understanding of our target audience. Through surveys, interviews, and focus groups, we gathered valuable insights into their needs, pain points, and expectations. This research helped us create user personas that served as guiding principles throughout the design process.
2. **Building the Information Architecture:** We meticulously designed the website's information architecture to ensure clarity and ease of navigation. The content was structured and organized in a logical manner, ensuring that users could find what they needed with minimal effort. Our aim was to create an intuitive experience where users could effortlessly navigate through the website, akin to traversing a well-planned city.
3. **Wireframing the Future:** To visualize the layout and functionality of key pages, we started with low-fidelity wireframes. These served as blueprints, allowing us to gather initial feedback and make iterative improvements before progressing to detailed visuals. This approach ensured that the UI design aligned with the intended user flow and goals.

4. **Where Beauty Meets Usability:** We carefully crafted a captivating UI that struck a balance between aesthetics and usability. Following user-friendly design principles, such as clarity, intuitiveness, and accessibility, we selected visuals that resonated with the target audience while conveying a sense of trust and professionalism. Our goal was to create an interface that users would find visually appealing and easy to engage with.
5. **Putting Users in the Driver's Seat:** Real users played a crucial role in shaping the UI and UX. We conducted extensive usability testing, observing how users interacted with the website and identifying any challenges they encountered. Through rigorous testing with representative users, we gathered valuable feedback that allowed us to fine-tune the UI and address potential issues. This user-centric approach ensured that the UI truly served its purpose and met the needs of our users.

Beyond the Basics:

In addition to the core UI and UX design, we incorporated several extra features to enhance the website's functionality and accessibility:

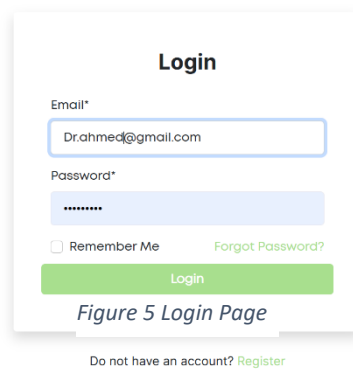
1. **Responsive Design:** The website seamlessly adapts to different devices, offering a flawless experience on desktops, tablets, and mobile devices.
2. **Accessibility Features:** We made sure the website caters to users with diverse needs, implementing accessibility features to ensure inclusivity and a positive experience for all users.
3. **Clear and Concise Language:** The website utilizes clear and concise language, avoiding jargon and technical terminology, ensuring that the content is easily understood by all users.

4.3.2. Front-end

The front-end of “NiceJob” is built on HTML, CSS, JavaScript, and the Laravel framework. These technologies, enhanced by the Bootstrap library, enable us to create reusable UI components for a consistent and easy-to-maintain design. The front-end communicates smoothly with the back-end through well-designed APIs, allowing for efficient data transfer and functionality. We test the front-end thoroughly on various devices and browsers, ensuring a flawless user experience for everyone. The front-end is the user’s gateway to “NiceJob”, providing an engaging and intuitive interface for their career exploration.

4.3.3. Website Pages

login Page

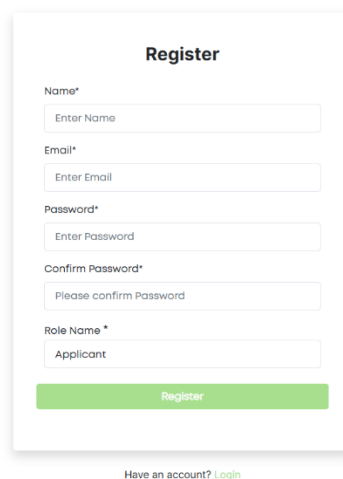


The login form is titled "Login" and is contained within a white box with a subtle shadow. It features two input fields: "Email*" with the placeholder "Dr.ahmed@gmail.com" and "Password*" with masked characters "*****". Below the password field are two links: "Remember Me" with an unchecked checkbox and "Forgot Password?". A green "Login" button is positioned below these links. At the bottom of the form box, the text "Do not have an account? Register" is displayed, with "Register" as a green link.

Figure 5 Login Page

This page displays a login form for users to enter their email and password.

Register page

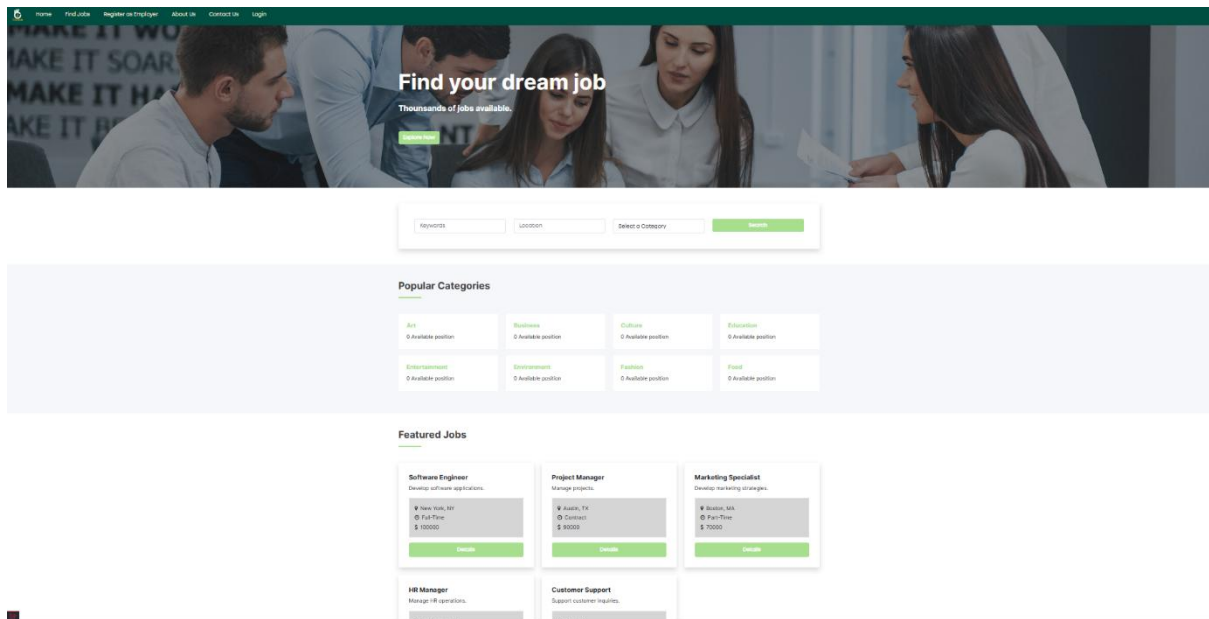


The register form is titled "Register" and is contained within a white box with a subtle shadow. It features five input fields: "Name*" with placeholder "Enter Name", "Email*" with placeholder "Enter Email", "Password*" with placeholder "Enter Password", "Confirm Password*" with placeholder "Please confirm Password", and "Role Name *" with placeholder "Applicant". A green "Register" button is located below the role name field. At the bottom of the form box, the text "Have an account? Login" is displayed, with "Login" as a green link.

Figure 6 Register Page

This page contains a registration form for new users on the website, allowing users to enter their name, email, password, confirm password, and select their role ("applicant," "employer," or "manpower consultant").

Home page



This webpage allows users to search for jobs based on their preferences and interests, featuring a search

Figure 7 Home Page

form, popular categories, featured jobs, and the latest jobs sections.

Personal Information page

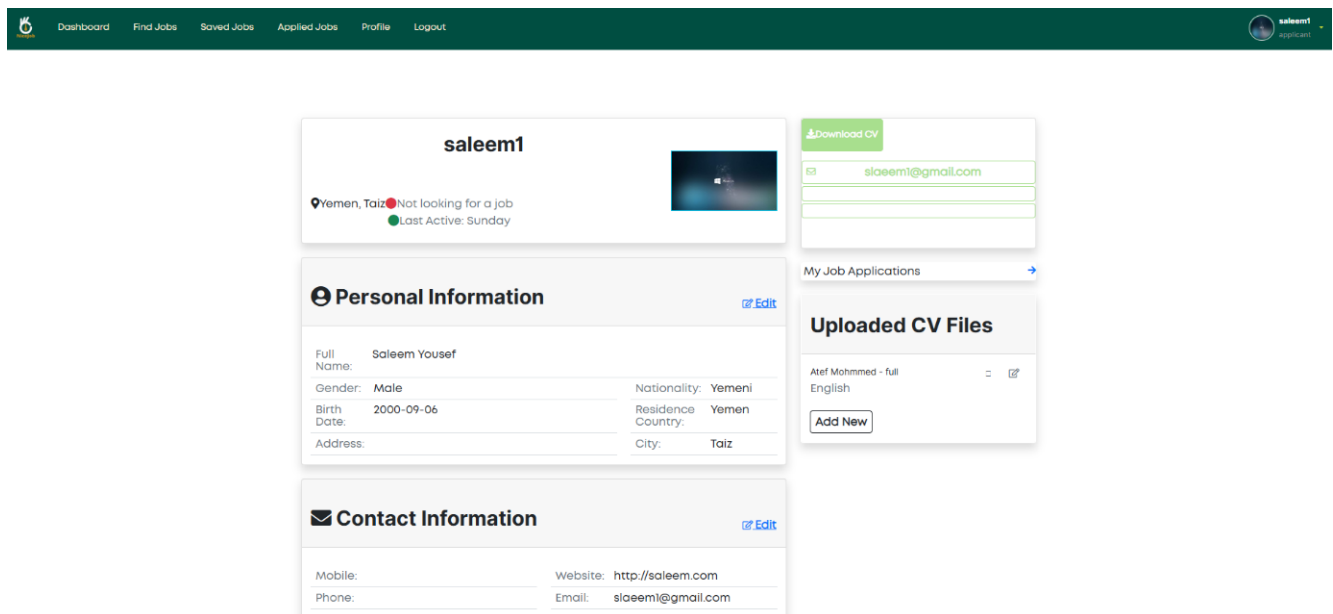


Figure 6 Personal Information Page

This page displays the user's personal information and contact details, including their name, designation, residence country and city, gender, nationality, address, mobile number, phone number, website, and email. The page also provides an option to edit the personal information

Create Jop page

The screenshot shows the 'Post a Job' form on the NiceJob.com dashboard. The form is titled 'Post a Job' and is located under the 'Account Settings' menu. It contains several sections for job details, company information, and job requirements.

Job Details

- Title ***: Input field for Job Title.
- Category ***: Dropdown menu for Select a Category.
- Job Type ***: Dropdown menu for Select Job Type.
- Vacancy ***: Input field for Vacancy.
- Salary**: Input field for Salary.
- Location ***: Input field for Location.
- Description ***: Rich text editor for job description.
- Desired Skills**: Rich text editor for desired skills.
- Qualifications**: Rich text editor for qualifications.
- Experience ***: Input field for Experience (e.g., 1 Year).
- Keywords**: Input field for keywords.

Company Details

- Name ***: Input field for Company Name.
- Location**: Input field for Location.
- Website**: Input field for Website.

Save Job: A green button to save the job posting.

Footer

About NiceJob.com
NiceJob Find your dream job in the modern world NiceJob is your ultimate job search platform. It connects you with thousands of employers, helps you showcase your skills, and gives you feedback and guidance. With NiceJob, you can take control of your career and explore the possibilities.

Quick Links
[About Us](#)
[Contact Us](#)

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Figure 9 Create A Job Page

This page contains a form for posting a job, with various input fields for job details, company details, and job requirements.

Job Details page

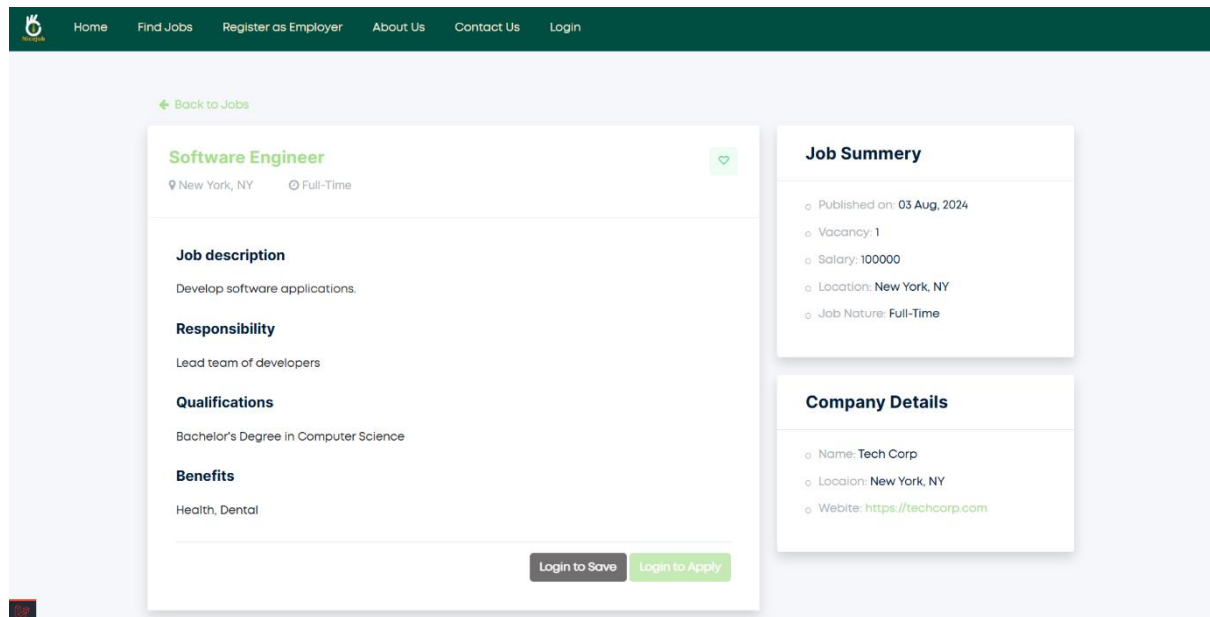


Figure 9 Job Details Page

This page displays the details of a job, showing job information, job summary, company details, and provides functionality to save and apply for the job.

Jops page

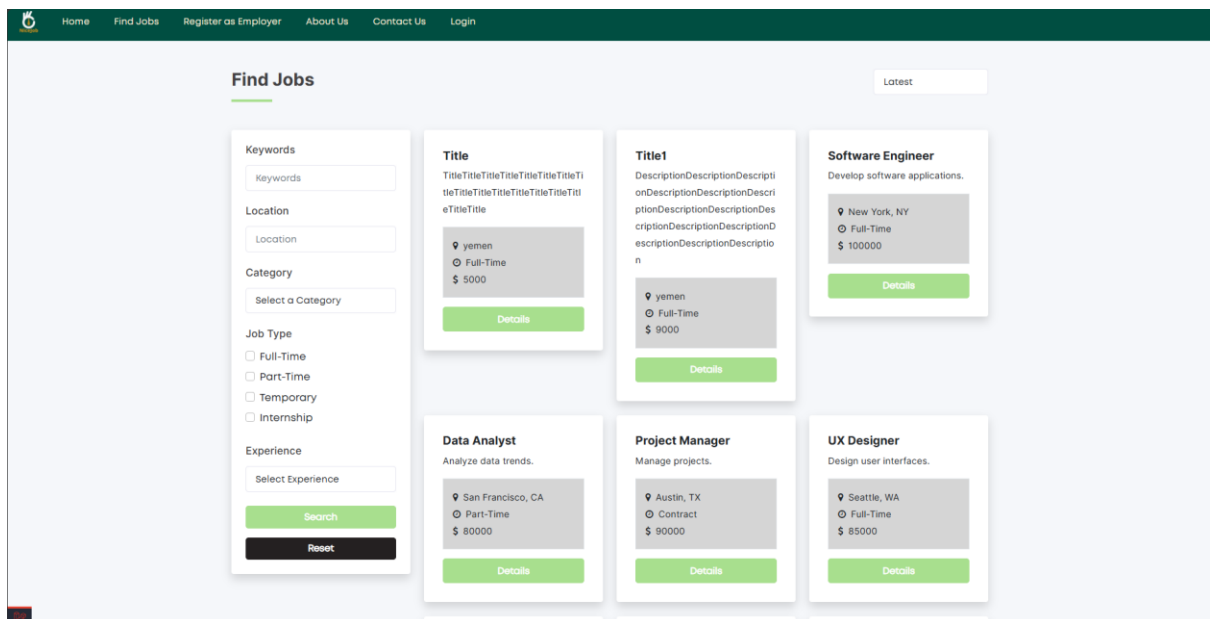


Figure 11 Jobs Page

This page is a job search page that allows users to enter keywords, location, category, experience, and job type to filter job listings. It also includes a sorting feature to sort the job listings based on the "Latest" or "Oldest" option.

Jop Applied page

Home / Account Settings

saleem1
Change Profile Picture

Account Settings
Jobs Applied
Saved Jobs
Logout

Jobs Applied

Title	Applied Date	Applicants	Status	Action
artist Part-Time - yemen	09 Aug, 2024	1 Applications	Active	

About NiceJob.com
NiceJob Find your dream job in the modern world NiceJob is your ultimate job search platform. It connects you with thousands of employers, helps you showcase your skills, and gives you feedback and guidance. With NiceJob, you can take control of your career and explore the possibilities.

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Figure 10Applied Job Page

This page displays the user's job applications in a table format, along with their application date, number of applicants, job status, and an action button with dropdown options for viewing or removing the job application. The page also includes pagination links for navigating through multiple pages of job applications.

4.3.4. Back-end

The back-end of “NiceJob” is powered by Laravel 10, a PHP framework that provides a robust and elegant solution for web development. Laravel 10 offers several advantages that make it an ideal choice for our project, such as:

1. **MVC Architecture:** Laravel 10 follows the Model-View-Controller (MVC) pattern, which separates the application logic from the presentation layer. This allows for a modular and maintainable codebase, as well as easier testing and debugging.
2. **Eloquent ORM:** Laravel 10 features Eloquent, an object-relational mapper (ORM) that simplifies the interaction with the database. Eloquent enables us to write expressive and fluent queries, as well as perform CRUD operations on our models. Eloquent also supports various database drivers, such as MySQL, PostgreSQL, SQLite, and MongoDB.
3. **Authentication and Authorization:** Laravel 10 provides a built-in authentication and authorization system that handles user registration, login, password reset, and role-based access control. We can easily customize and extend this system to meet our specific requirements, such as adding social login, email verification, and multi-factor authentication.
4. **Blade Templating Engine:** Laravel 10 uses Blade, a powerful and flexible templating engine that allows us to write dynamic HTML views. Blade supports inheritance, components, directives, and other features that enhance the readability and reusability of our views. Blade also compiles to plain PHP, which ensures fast performance and compatibility.
5. **Artisan Console:** Laravel 10 comes with Artisan, a command-line interface that provides a range of useful commands for developing and managing our application. We can use Artisan to generate code scaffolding, run migrations, seed the database, run tests, and more. We can also create our own custom commands to automate common tasks.

Database Implementation:

The database of “NiceJob” is implemented using MySQL, a relational database management system that is widely used and supported by Laravel 10. MySQL offers several benefits for our project, such as:

1. **Scalability:** MySQL can handle large amounts of data and support concurrent users without compromising performance or reliability.
2. **Security:** MySQL provides various features to ensure the security and integrity of our data, such as encryption, authentication, authorization, and backup.
3. **Flexibility:** MySQL allows us to define and modify the structure and schema of our data as needed, as well as support various data types and formats.

4. Compatibility: MySQL is compatible with various operating systems, platforms, and languages, making it easy to integrate with our front-end and back-end technologies.

We used Laravel's migration and seeding features to create and populate our database tables. Migration is a way of creating and modifying database tables using PHP code, which makes it easier to track and manage changes to the database structure. Seeding is a way of inserting dummy data into the database tables, which is useful for testing and development purposes.

We also used Laravel's Eloquent ORM to interact with our database. Eloquent is an object-relational mapper that maps our database tables to corresponding PHP classes, called models. This allows us to perform CRUD (create, read, update, delete) operations on our data using simple and expressive syntax, without writing raw SQL queries. Eloquent also supports various features that make working with our data more convenient and efficient, such as:

1. Relationships: Eloquent allows us to define and access the relationships between our models, such as one-to-one, one-to-many, many-to-many, and polymorphic relationships. This enables us to query and manipulate related data with ease.
2. Scopes: Eloquent allows us to define and apply custom filters to our queries, called scopes. This helps us to reuse common query logic and simplify our code.
3. Mutators and Accessors: Eloquent allows us to modify and format the attributes of our models before saving them to the database or retrieving them from the database, called mutators and accessors. This helps us to ensure the consistency and validity of our data.
4. Events and Observers: Eloquent allows us to listen and respond to various events that occur when working with our models, such as creating, updating, deleting, and restoring. This enables us to perform additional actions or logic when these events happen, such as sending notifications, logging activities, or validating data. We can use observers to group the event listeners for a given model in a single class, which makes our code more organized and maintainable.

1.3. Conclusion

This chapter summarizes the transformation of "NiceJob" from vision to reality. Our platform meets the needs of job seekers and employers with user-centric design, Laravel, and rigorous testing. We've created an attractive, intuitive, and accessible user interface, along with a robust and scalable back-end. But this is just the beginning. In the upcoming chapters, we'll unveil the unique features that make "NiceJob" an innovative tool for connecting people and opportunities. We'll also discuss how we can adapt to the dynamic job market.

CHAPTER FIVE: TESTING

5.1 Introduction

Testing is a critical phase in the software development process, where the application is rigorously evaluated to ensure it meets the required standards of quality. In this chapter, we will delve into the various testing methodologies applied to "NiceJob", emphasizing the systematic approach to identifying and resolving issues. The objective of this phase is not only to validate the functionality and performance of the platform but also to guarantee its reliability and security in the real world.

The testing process for our platform is meticulously designed to cover all aspects of the system, from individual units of code to the overall user experience. This ensures that every component operates as intended and interacts harmoniously with others. By the end of this chapter, we aim to demonstrate the thoroughness of our testing procedures and the robustness of the final product, ready for deployment and use by job seekers and employers alike.

5.2 Steps Of Software Testing

The software testing process for "NiceJob" is an extensive, layered approach that guarantees the delivery of a superior product. Our methodology is carefully organized into distinct phases, each with specific objectives and tasks. The steps below delineate our systematic method for affirming the platform's functionality, performance, and security:

1. **Requirement Analysis:** This foundational phase involves an in-depth review of the software requirements to establish a clear understanding of the anticipated system behavior. It is a pivotal step that lays the groundwork for all subsequent testing activities.
2. **Test Planning:** In this strategic phase, we formulate a test plan that defines the testing objectives, resources, schedule, and deliverables. It acts as a blueprint for the testing process and ensures that all team members are in sync with the testing objectives.
3. **Test Case Development:** Drawing from the test plan, we create a collection of test cases that cover a range of input scenarios and user behaviors. Each test case is tailored to verify particular facets of the system and includes predefined conditions and expected results.
4. **Test Environment Setup:** We set up a controlled testing environment that mirrors the production setting to ensure that our tests produce precise outcomes. This setup involves configuring hardware, software, network settings, and other essential components.
5. **Test Execution:** Once the environment is ready, we methodically carry out the test cases, carefully recording any variances between the actual and expected results. These variances are identified as defects and undergo further examination.

6. Defect Tracking: All discovered defects are recorded in a tracking system, which enables us to oversee their status, delegate resolution responsibilities, and rank them based on severity and impact.
7. Re-testing and Regression Testing: Following the resolution of defects, we perform re-testing to verify their rectification. Additionally, regression testing is conducted to ensure that recent modifications have not negatively impacted existing functionalities.
8. Performance Testing: This phase evaluates the system's conduct under different load conditions. It is vital for pinpointing potential constraints and confirming that the application can manage the anticipated user load.
9. User Acceptance Testing (UAT): We involve a select group of end-users to authenticate the system against their requirements. UAT represents the ultimate validation step before the software is considered ready for launch.

These steps are carried out with meticulous precision and thoroughness, ensuring that "NiceJob" not only fulfills but surpasses the expectations of its users and stakeholders.

5.3 Testing Requirements

For a Laravel-based website, the testing environment must be carefully curated to ensure that it supports the Laravel framework and the technologies it interacts with. The following are the specific hardware and software requirements, along with the recommended tools for a comprehensive testing strategy:

Hardware Requirements

A development machine with a multi-core processor, such as an Intel i5 or i7, or equivalent, to efficiently handle the Laravel framework and associated services. A minimum of 8GB RAM to ensure smooth multitasking and to support the execution of the application along with testing tools. At least 256GB of SSD storage to provide quick read/write access for the application, databases, and test artifacts. A stable and fast internet connection to test the application under realistic network conditions and to access cloud-based testing services.

Software Requirements

The latest stable release of the Laravel framework to take advantage of the newest features and security updates. A compatible operating system, such as the latest stable release of Ubuntu for Linux or the most recent version of macOS or Windows, depending on the development preference. Development tools like PHPStorm or Visual Studio Code that offer Laravel support for efficient coding and debugging. All necessary PHP extensions and Composer dependencies required by Laravel and the project's specific packages.

Testing Tools

PHPUnit for automated unit and feature testing, as it is natively supported by Laravel. Laravel Dusk for browser automation and application testing from a user's perspective. Postman for API testing, ensuring that all endpoints function correctly and efficiently. Git for version control, integrated with a CI/CD pipeline that includes running tests on code push or pull requests. Laravel Telescope for debugging and monitoring your application during the testing phase. JIRA or Trello for defect tracking and project management, allowing for organized tracking of issues and collaborative resolution.

5.4 Testing Cases

In the pursuit of excellence for our platform we have devised a series of sophisticated testing scenarios that rigorously evaluate the platform's capabilities. These scenarios are systematically classified into distinct categories, each designed to scrutinize a specific aspect of the system's functionality and security. The categories include functional, performance, security, and user acceptance testing. Each testing case is meticulously documented, detailing the initial conditions, the procedural steps to be executed, and the anticipated outcomes. Below are exemplar cases from each category:

Functional Test Case:

Objective: Validate the operational integrity of the login mechanism.

Procedure: Enter valid user credentials and initiate the login process.

Expected Outcome: Successful authentication and access to user-specific features.

Performance Test Case:

Objective: Evaluate the system's responsiveness under high-demand scenarios.

Procedure: Simulate peak user load and measure response times for critical operations.

Expected Outcome: Consistent performance that meets predefined benchmarks for speed and efficiency.

Security Test Case:

Objective: Ascertain the resilience of the search functionality against SQL injection attacks.

Procedure: Execute a series of crafted inputs designed to exploit SQL vulnerabilities.

Expected Outcome: No unauthorized data access or system behavior alterations.

User Acceptance Test Case:

Objective: Confirm that the job search module aligns with end-user expectations.

Procedure: Engage real users to interact with the job search feature and collect feedback.

Expected Outcome: Positive user feedback indicating satisfaction with the feature's functionality and usability.

These testing cases are the cornerstone of our quality assurance process, ensuring that "NiceJob" is not only robust and secure but also aligned with the needs and expectations of its users.

5.5 Conclusion

The testing chapter culminates with a strong affirmation of the platform's readiness for deployment. The extensive and thorough testing procedures have been pivotal in refining the system, ensuring its functionality, security, and user experience align with the highest standards.

This phase has been crucial in identifying areas for improvement, fortifying the system's resilience, and confirming its capability to meet the demands of its intended users. The dedication to quality exhibited here is reflective of the team's commitment to excellence and their focus on delivering a product that users can rely on.

As we move towards deployment, the insights gained from the testing process will continue to guide ongoing enhancements. The foundation laid by this rigorous testing ensures a robust launch and a promising future for the platform, poised to become an essential tool in its respective domain.

Chapter Six: Conclusion and Future Work

6.1 Conclusion

"NiceJob" is a testament to the power of technology in bridging job market gaps. Meticulously crafted with the user experience in mind, it simplifies job hunting and recruitment. Through testing and innovation, the platform ensures a seamless and secure experience. With future expansion and new features, "NiceJob" will remain an indispensable tool for job seekers and employers. Its release marks a meaningful impact on the job market ecosystem.

6.2 Future Work

As we look ahead, "NiceJob" holds immense potential for growth and continuous improvement. The following areas represent exciting opportunities for enhancing the platform:

1. Mobile Application Development:
 - Extend the platform's reach by creating a mobile app.
 - Enable users to search for jobs conveniently on their smartphones.
2. Global Expansion:
 - Adapt the platform for international job markets.
 - Support multiple languages and comply with diverse employment regulations.
3. Community Building:
 - Foster a sense of community within the platform.
 - Create forums and networking opportunities for professionals and industry experts.
4. Continuous Learning Resources:
 - Offer integrated learning modules.
 - Help job seekers acquire new skills and certifications relevant to their career goals.

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